

ViMech



Mechanical valves: fluoroscopic identification and design for valve-in-valve

User manual and complete catalogue



https://hcuvicicor.github.io/ViMech_app/

Data version 2026-10-06 · October 8, 2026

9 brands · 36 models · 5 languages

ICICOR — Instituto de Ciencias del Corazón · Hospital Clínico Universitario de Valladolid

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1. Bicarbon™ — Overline Aortic (supra-annular), Slimline Aortic, Fitline Aortic, Fitline Mitral
2. CarboMedics — Standard Aortic, Reduced Aortic, Top Hat Aortic (totally supra-annular), Orbis Aortic, Standard Mitral, Optiform Mitral, Orbis Mitral
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4. HP-Biplus — Standard Aortic, HP-Biplus C
5. Jyros — Standard Aortic
6. Medtronic Open Pivot™ (ATS) — Open Pivot™ Standard Aortic (intra-annular), Open Pivot™ AP / AP360™ Aortic (supra-annular), Open Pivot™ Standard Mitral, Open Pivot™ AP360™ Mitral (supra-annular)
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1 Introduction

ViMech is a free web application developed by ICICOR (Instituto de Ciencias del Corazón, Hospital Clínico Universitario de Valladolid). It brings together the fluoroscopic identification, design, dimensions and TAVI sizing of bileaflet mechanical heart valves, as a reference for *valve-in-mechanical* procedures (TAVI inside a mechanical prosthesis).

This manual contains the complete guide to every function, the full catalogue of the 9 brands and 36 models with their photographs, dimensions, TAVI tables and sources, and the licence, rights and credits information.

Intended purpose: ViMech is an educational reference that gathers published dimensions of mechanical heart valves and fluoroscopic identification features. It does not process patient data, does not make diagnostic or therapeutic decisions and is not a medical device.

This tool is for informational and educational purposes only. It does not replace the manufacturer's instructions for use, imaging-based assessment or clinical judgement. Always verify dimensions against the official documentation before any procedure.

TAVI sizing suggestions are indicative and must be confirmed by the heart team. The creators accept no liability for errors or omissions.

Non-affiliation: this is an independent application, not affiliated with, endorsed or sponsored by any of the manufacturers mentioned.

Access, installation and offline use

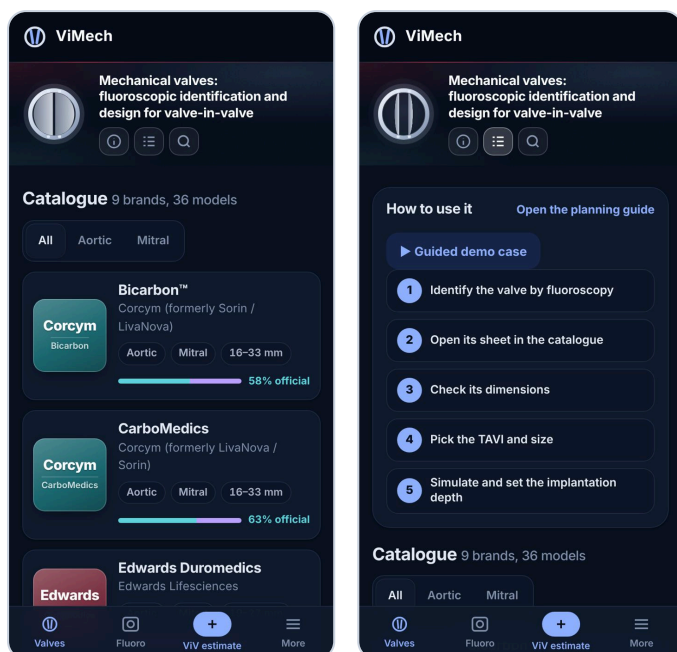
- Address: https://hcuvicicor.github.io/ViMech_app/ (also through the QR code on the cover).
- It works on phone, tablet and computer, in any current browser. It requires no registration and collects no patient data.
- It can be installed as an app: on Android/Chrome, menu : → "Install app"; on iPhone/Safari, Share → "Add to Home Screen".
- Once opened, it keeps working offline and updates itself when a new version is available.
- Languages: Spanish, English, Portuguese, French and Italian (More → Language). Themes: dark (default), grey, aurora, light and automatic (More → Theme).

2 Function guide

Home screen and catalogue

The app opens on the catalogue of brands. Each card shows the manufacturer, the available positions (aortic/mitral), the size range and the percentage of values from an official source.

- ⓘ shows the app description.
- ☰ **How to use it** opens the 5 workflow steps, the link to the planning guide and the guided demo case.
- 🔍 opens the search by brand, model or catalogue code.
- The **All / Aortic / Mitral** filters limit the catalogue to one position.
- Bottom bar: Valves, Fluoro, ViV estimate and More (menu).



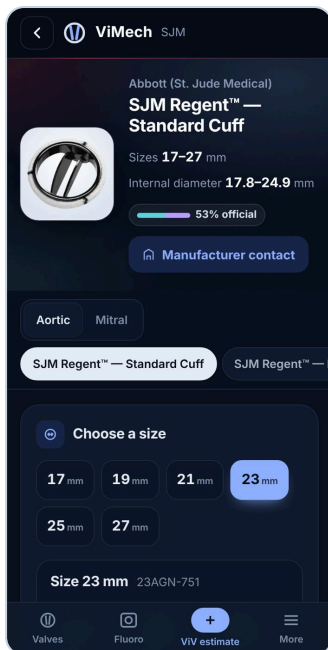
Catalogue

How to use it

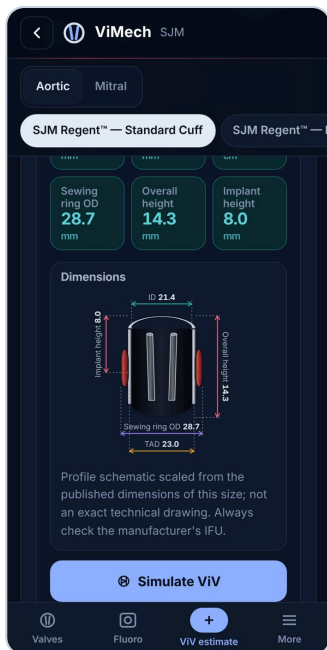
Valve sheet

Each brand has a sheet with a position selector (aortic/mitral) and a model selector. It contains:

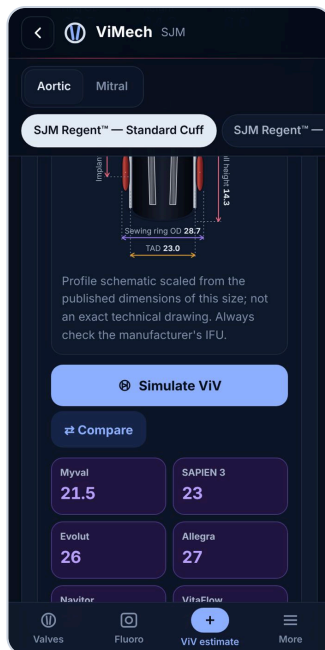
- Images and design (tap to enlarge; pinch to zoom).
- **Choose a size:** tapping a size shows its key dimensions (ID, TAD, GOA, EOA, heights), the dimension schematic to scale and the suggested TAVI sizes for each device. Amber values with ⚠ mean "Caution".
- **Simulate ViV** and **Compare** buttons.
- Full technical specification table; tap a row to open that size.
- Fluoroscopic identification, positioning and deployment, and the source of every value with its colour code: 🟢 official source, 🟡 unofficial (literature, distributors), 🟣 hospital own data.
- **Manufacturer contact** (public phone, email, address and website) and **Report an error** (opens an email with the sheet details).



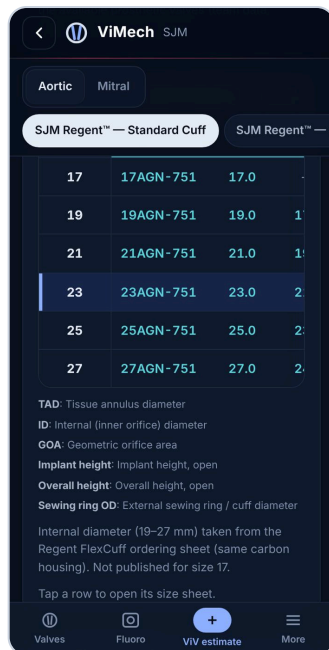
Sheet header



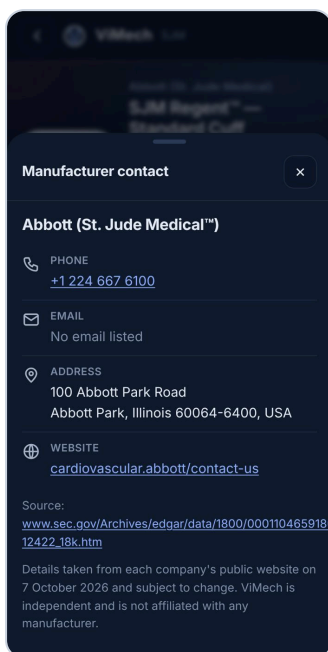
Selected size



Dimension schematic



Specification table



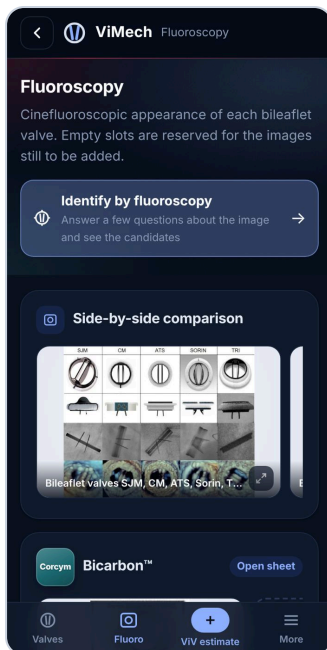
Manufacturer contact

Fluoroscopy and identification

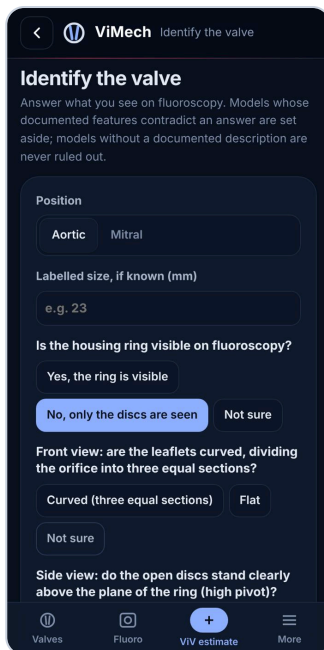
The **Fluoro** tab gathers the cinefluoroscopic atlas of all valves. From there, from the planning guide or from the menu you can open the **Identify by fluoroscopy** assistant:

- Enter the position and, if known, the labelled size.
- Answer what you see on the image: whether the ring is visible, whether the discs are curved, whether the pivot is high, whether there is a second denser ring and whether the discs are very dense.
- Models whose documented features contradict an answer are set aside; models without a documented description are never ruled out. Candidates are grouped by brand with their description, images and a link to each sheet.

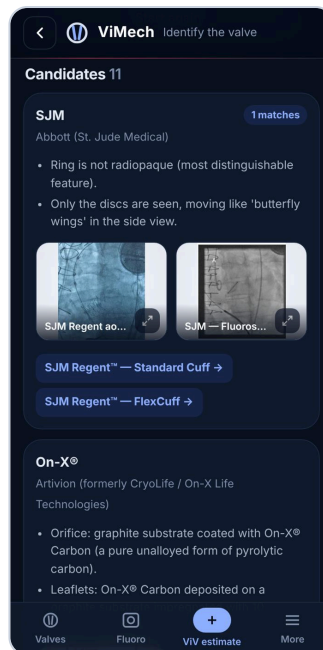
Orientative aid built only from the fluoroscopic descriptions in the app (ViMech article). Always confirm against the images, the patient's records and the manufacturer's documentation.



Fluoroscopy atlas



Questions

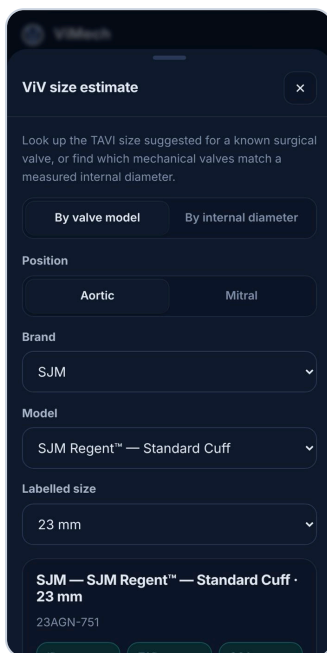


Candidates

ViV estimate

The central bottom-bar button opens the size estimate, with two modes:

- **By valve model:** choose position, brand, model and size to see the TAVI sizes from the table, with access to the full sheet.
- **By internal diameter:** enter a measured ID and a tolerance to list the valves whose published ID matches, ordered by closeness, with the closest highlighted.



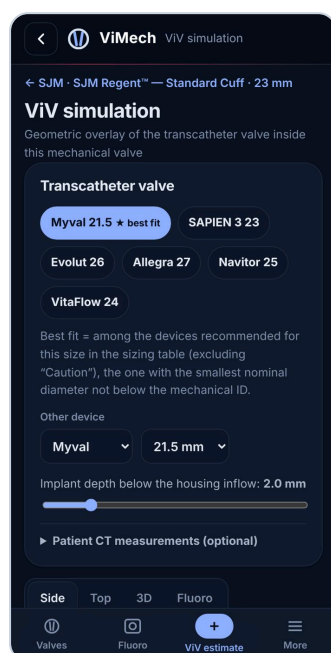
Estimate by model

ViV simulation

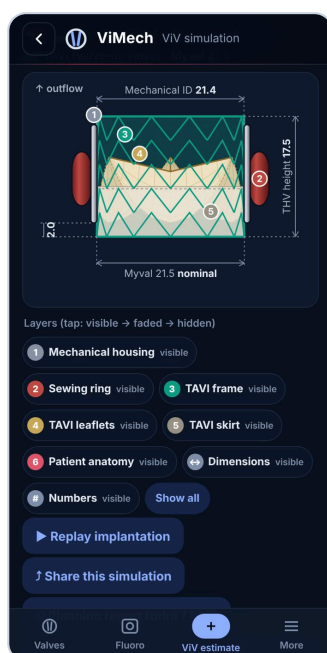
From any size sheet, **Simulate ViV** overlays the transcatheter valve (TAVI) to scale inside the already implanted mechanical prosthesis. On opening, the TAVI arrives crimped from the aorta and expands inside the ring (**Replay implantation** button).

- **TAVI selection:** the best fit is proposed automatically (Best fit = among the devices recommended for this size in the sizing table (excluding "Caution"), the one with the smallest nominal diameter not below the mechanical ID.); the other table options appear as buttons and any other device and size can be chosen.
- **Implant depth:** slider that lowers or raises the TAVI relative to the ring inflow.
- **Views:** Side (to-scale section with dimensions), Top (diameter inside the ring vs nominal), 3D (drag to rotate) and Fluoro (simulated radiological projection, with tilt, rotation and coplanar-view indicator).
- **Layers:** mechanical housing, sewing ring, TAVI frame, leaflets and skirt, patient anatomy, dimensions and numbers; each tap goes visible → faded → hidden.
- **Results:** mechanical ID, nominal diameter, oversizing vs ID, diameter inside the ring, constraint, height and portion above the ring; warnings if the combination is "Caution" or the TAVI is smaller than the ID.
- **Patient CT measurements** (optional): height of each coronary, sinus Ø, STJ Ø and height. The aortic root is drawn, the frame is limited by it and the THV–coronary distance is computed at the height of each ostium.
- **Candidate comparison:** table with oversizing and constraint of each option; tap a row to simulate it.
- **Share this simulation:** creates a link that keeps device, size, depth, view and CT measurements.
- **Planning report (print / PDF):** A4 sheet with the valve, the TAVI, results, comparison, drawings and sources, with a blank line to write the case by hand.

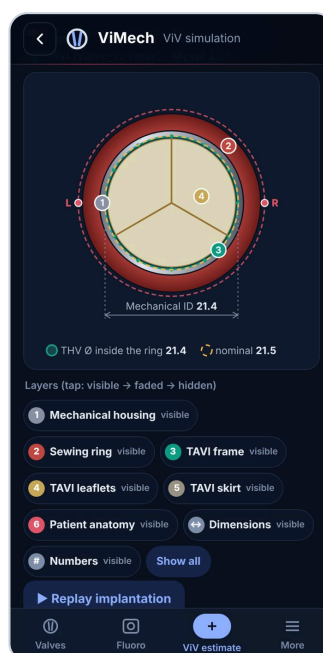
Orientative geometric simulation built from nominal dimensions. It does not model the discs/leaflets or pivots of the mechanical valve, tissue, calcium, frame recoil or the actual expansion, and does not replace CT-based planning or the manufacturers' IFU. The narrowing at the edges of the ring is drawn as a gradual transition of schematic length.



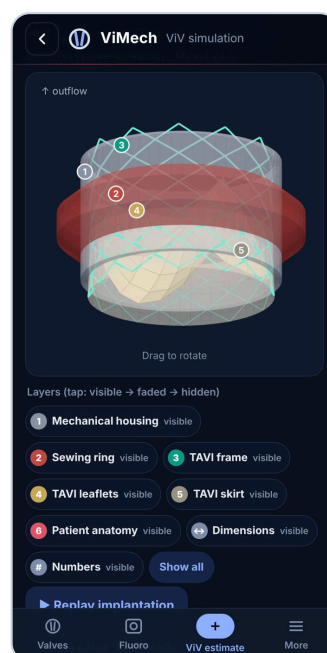
TAVI selection



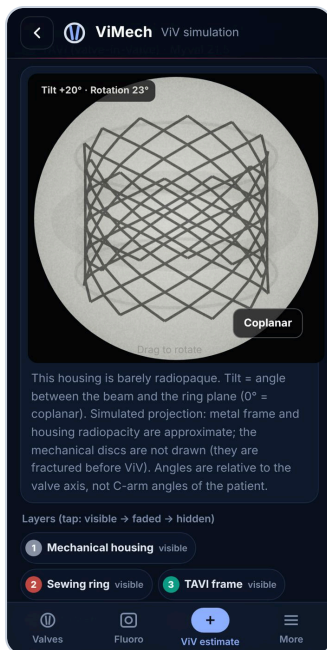
Side view



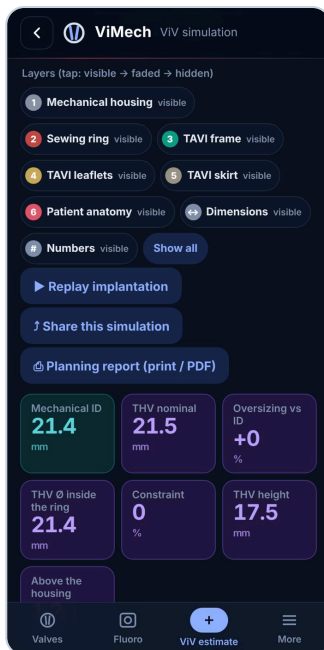
Top view



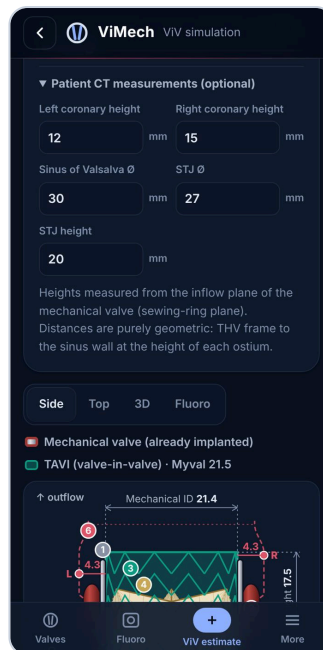
3D view



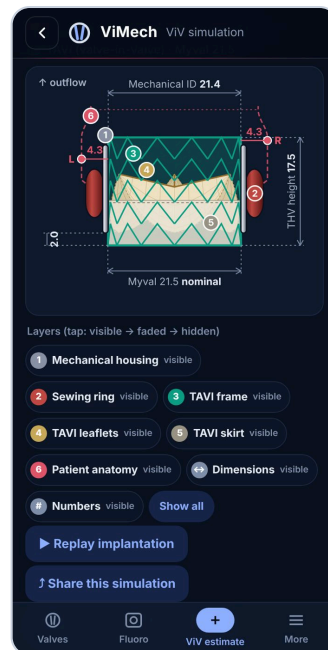
Simulated fluoroscopy



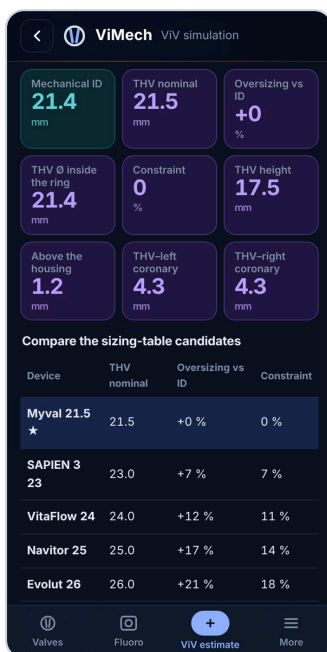
Layers



CT measurements



Root and coronaries



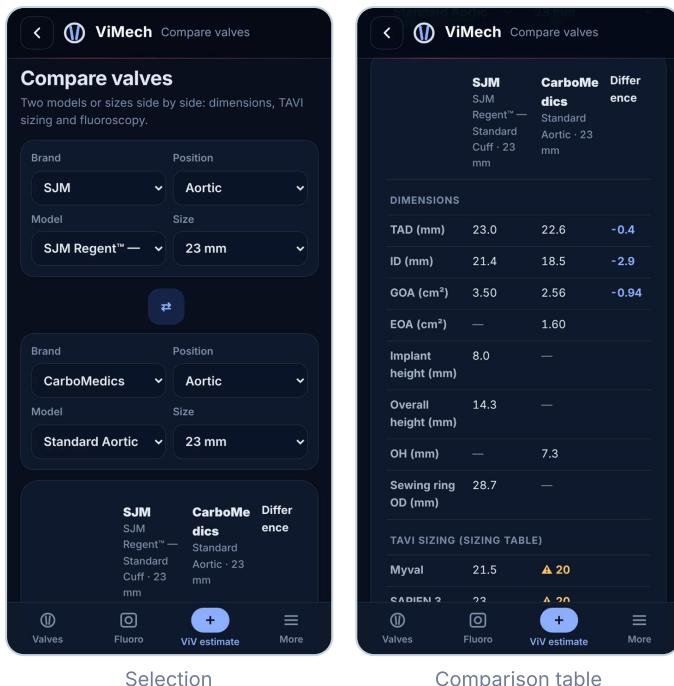
Results



Comparison

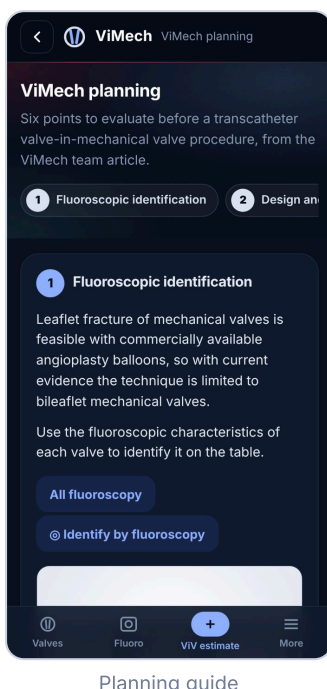
Compare valves

Puts two models or sizes side by side (from “⇌ Compare” on the sheet or from the menu): all their dimensions with the difference, the TAVI sizes from the table and the design and fluoroscopy images. The ⇌ button swaps both sides.



Planning guide

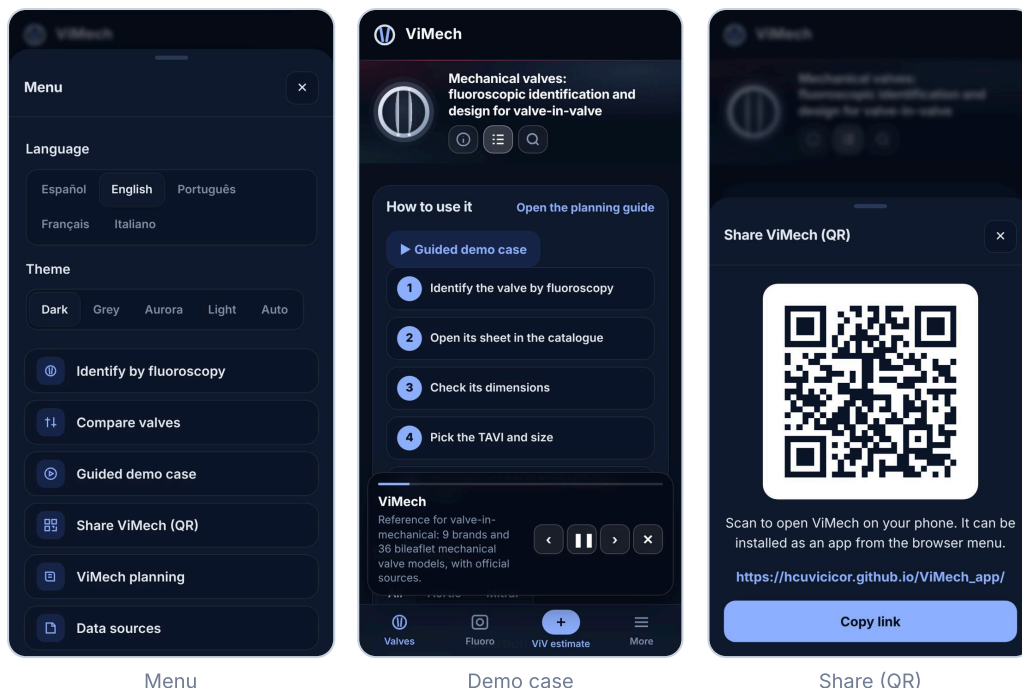
Six points to evaluate before a valve-in-mechanical procedure, from the ViMech team article, with figures and references (reproduced in full in the next chapter).



Planning guide

Menu, demo case and sharing

- **More** opens the menu: language, theme, identify by fluoroscopy, compare valves, guided demo case, share ViMech (QR), planning guide, data sources, about, disclaimer, licence and credits, contact, privacy policy and this manual.
- **Guided demo case**: walks through the whole workflow by itself in 9 steps of about 7 s with the SJM Regent 23 mm; bottom bar with previous, pause, next and exit. The address `.../ViMech_app/?demo=loop` repeats it in a loop (useful for congress screens).
- **Share ViMech (QR)**: QR code and link to open the app on another device.



Menu

Demo case

Share (QR)

Abbreviations

Size	Size (mm)
TAD	Tissue annulus diameter
Outer dia.	Outer diameter
ID	Internal (inner orifice) diameter
GOA	Geometric orifice area
Orifice area	Internal orifice area
EOA	In vivo effective orifice area (manufacturer)
Implant height	Implant height, open
Overall height	Overall height, open
OH	OH as listed by the manufacturer
Overall profile	Overall profile height
Inflow profile	Inflow profile height
Outflow profile	Outflow profile height
Sewing ring OD	External sewing ring / cuff diameter
Profile h closed	Profile height, closed
Profile H open	Profile height, open
THV / TAVI	Transcatheter heart valve
ViV	Valve-in-valve
IFU	Manufacturer's instructions for use

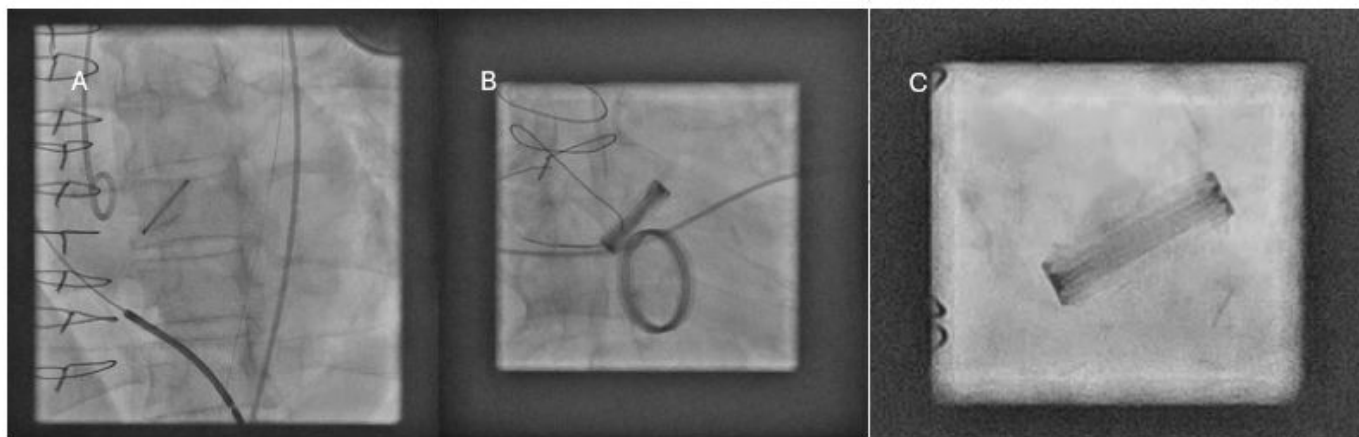
3 ViMech planning guide

Six points to evaluate before a transcatheter valve-in-mechanical valve procedure, from the ViMech team article.

1. Fluoroscopic identification

Leaflet fracture of mechanical valves is feasible with commercially available angioplasty balloons, so with current evidence the technique is limited to bileaflet mechanical valves.

Use the fluoroscopic characteristics of each valve to identify it on the table.



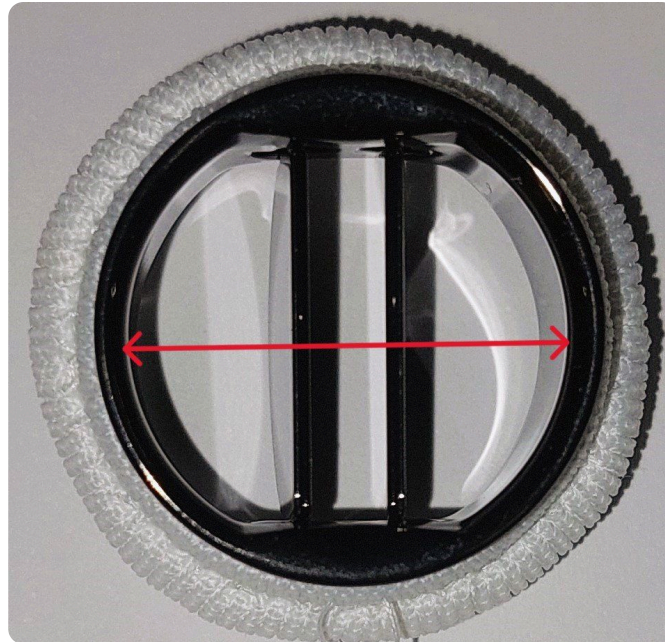
Fluoroscopic identification: A St Jude, B CarboMedics, C ATS.

2. Design and true internal diameter

The nominal size of a bileaflet valve is broadly consistent with the tissue annulus it is meant to fit. The measurements that matter for TAVI sizing are tissue annulus diameter, geometric orifice area and internal diameter.

The true internal diameter (ID) is not the tissue annulus diameter (TAD). TAD is the smallest flow area within the native annulus, measured intraoperatively with cylindrical sizers; ID is the minimum diameter inside the prosthesis through which blood flows.

Size the THV on the true ID and the geometric orifice area. Rule out pannus with transoesophageal echocardiography, as it can reduce the true ID further.



Internal diameter of a mechanical valve.

3. THV size and design

Know the design of the THV (intra-annular or supra-annular) to position it correctly inside the prosthesis.

Balloon-expandable valves are generally preferred for their radial strength and their ability to fracture or entrap a disc that has not dislodged with predilation.

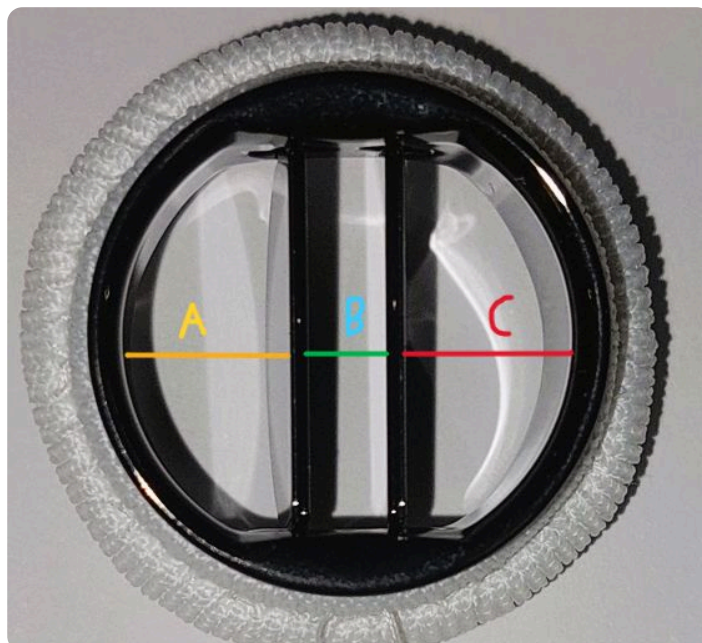
Each valve sheet shows the suggested size of every commercial THV for each prosthesis size.

4. Wire crossing and predilation

Bench testing showed the wire should cross between the two leaflets (orifice B), which facilitates leaflet fracture during predilation.

Balloon predilation was the most effective way to dislodge the discs, more than snaring. Use a peripheral balloon sized on CT and fluoroscopy measurements.

One or both discs may fracture and embolise. Avoid rapid pacing during predilation so the embolised discs do not move into the left ventricle.



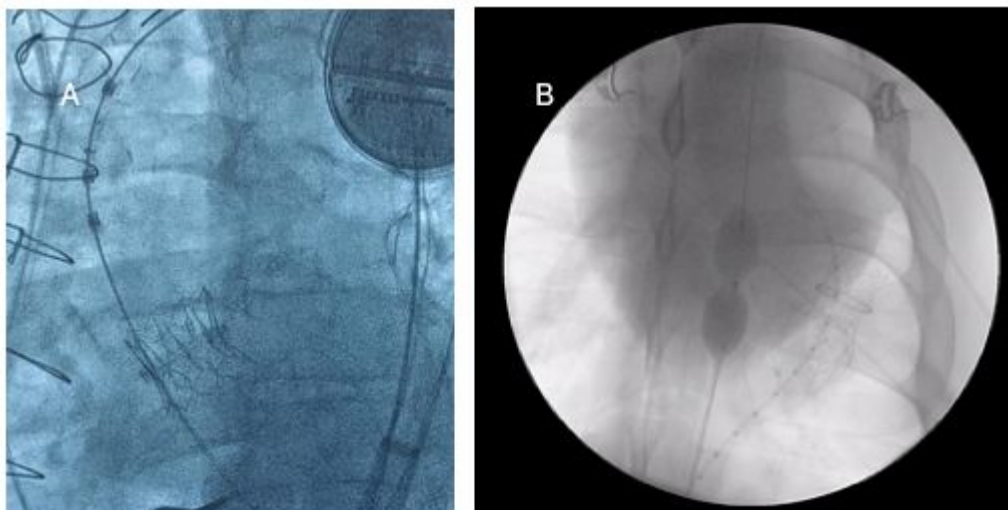
Cross the wire through orifice B to facilitate leaflet fracture during predilation.

5. Implantation position

The prosthetic valve ring is the landmark for THV implantation. Orient the valve in a lateral view on fluoroscopy.

Balloon-expandable valves: ideal height 80:20 (aortic:ventricular). Self-expanding valves: depth up to 3 mm is acceptable.

A higher implant can lead to valve embolisation.

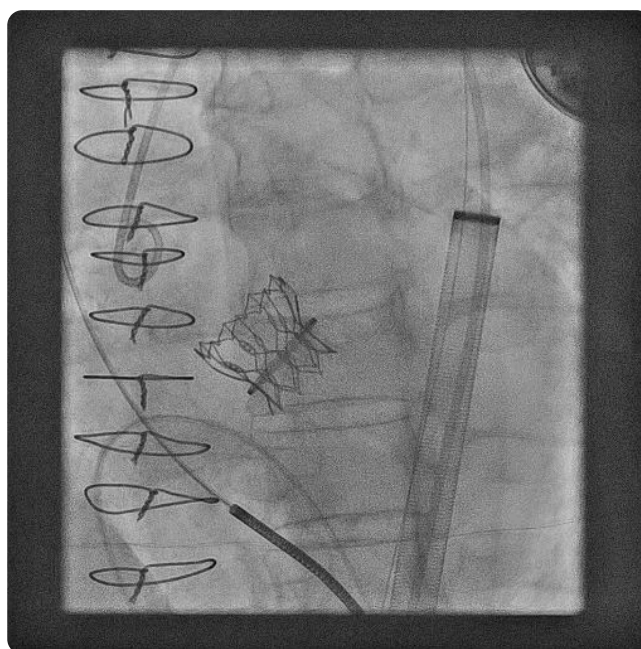


Implantation depth for TAVI in mechanical valves: A BEV in SJM, B SEV in SJM. Video 1: Myval implantation in a St Jude valve.

6. Anticipating complications

Small mechanical valve: consider TAVI with caution, as current THVs may cause patient-prosthesis mismatch and underexpansion because of the small internal diameter. The team implants slightly higher and flares the BEV.

Disc fragments: cerebral protection with a Flower or Sentinel device is advised. Leave the distal end of the Flower device inside the 22F sheath to avoid distal embolisation. In the team's experience, fragments reaching the peripheral circulation have endothelialised safely.



BEV flaring in small mechanical valves to avoid patient-prosthesis mismatch.

Key references

1. Amat-Santos IJ, Real C, Galán-Arriola C, et al. Transcatheter aortic valve-in-mechanical valve replacement: a first-in-human study. *Eur Heart J*. 2026. doi:10.1093/eurheartj/ehag019
 2. Jankowska PA, Butter C, Kühnel RU. Serial in vitro investigation of the fracture properties of mechanical heart valves. *JACC Cardiovasc Interv*. 2026;19(4):505-513. doi:10.1016/j.jcin.2025.11.040
 3. Kalçık M, Güner A, Yesin M, et al. Identification of mechanical prosthetic heart valves based on distinctive cinefluoroscopic and echocardiographic markers. *Int J Artif Organs*. 2019;42(11):603-610. doi:10.1177/0391398819860845
 4. Chambers JB, Oo L, Narracott A, et al. Nominal size in six bileaflet mechanical aortic valves: a comparison of orifice size and biologic equivalence. *J Thorac Cardiovasc Surg*. 2003;125(6):1388-93. doi:10.1016/s0022-5223(03)00039-4
- Content from the ViMech team article (manuscript in preparation).

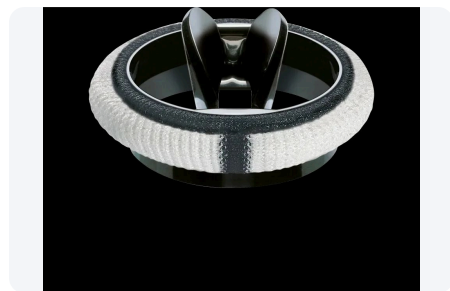
4 Valve catalogue

For each brand and model: photographs, fluoroscopic identification, positioning, full dimension table, TAVI size table and sources. Provenance colours: ■ Official source · ■ Unofficial data (literature, distributors) · ■ Hospital own data.

Bicarbon™

Corcym (formerly Sorin / LivaNova)

Overline Aortic (supra-annular) Aortic



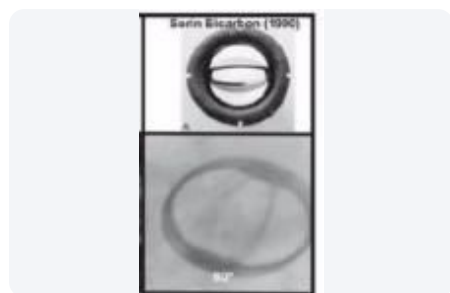
Sorin Bicarbon aortic — Photo · © Corcym



Sorin Bicarbon family — Photo · © Corcym



Sorin Bicarbon Overline aortic — Photo · © Corcym



Sorin Bicarbon 1990 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- The ring and leaflets are radiopaque.
- Frontal view: its distinctive characteristic is a curved profile (which separates the valve orifice into three equivalent sections).
- Side view: pathognomonic hazy appearance of the curved leaflets.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
16	ICV0870	15.2	15.2	1.76	0.97	6.0
18	ICV0871	17.2	17.2	2.27	1.54	6.4
20	ICV0872	19.2	19.2	2.83	2.07	6.8
22	ICV0873	21.3	21.3	3.45	2.39	7.2
24	ICV0874	23.3	23.3	4.14	3.06	7.6

TAVI SIZE SELECTION

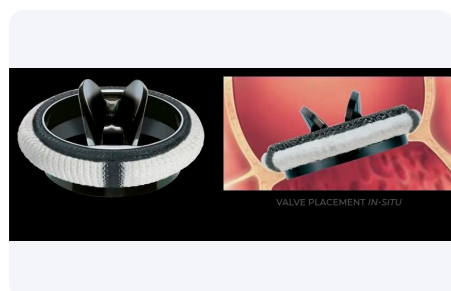
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
16	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
18	▲ Caution	▲ Caution	▲ Caution	23	▲ Caution	21
20	20/21.5	20	23	23	23	21
22	21.5	23	26	23	25	24
24	23/24.5	23	26	27	25/27	24/27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Corcym — Bicarbon™ Family brochure CC-MK-00002 A — <https://www.corcym.com/devices/aortic>
- Own data of the hospital research unit (ICICOR).

Slimline Aortic Aortic



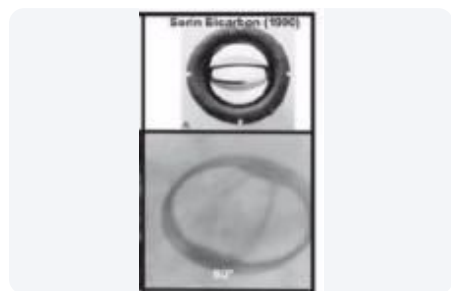
Sorin Bicarbon Slimline aortic — In-situ placement
· © Corcym



Sorin Bicarbon Slimline aortic — Photo · © Corcym



Sorin Bicarbon Slimline aortic — In-situ placement
· © Corcym



Sorin Bicarbon 1990 — Photo and fluoroscopy

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17	ICV0934	17.2	15.2	1.76	1.01	6.0
19	ICV0935	19.2	17.2	2.27	1.50	6.4
21	ICV0936	21.3	19.2	2.83	1.90	6.8

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
23	ICV0937	23.4	21.3	3.45	2.39	7.2
25	ICV0938	25.6	23.3	4.14	3.06	7.6
27	ICV0939	28.0	25.6	5.00	3.45	8.0

TAVI SIZE SELECTION

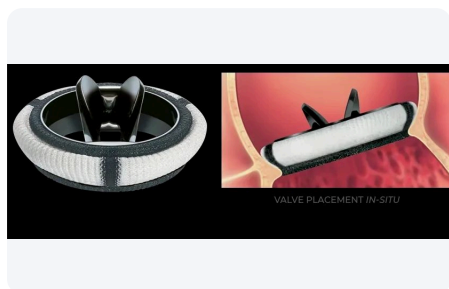
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
17	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
19	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	23	25	24
25	23/24.5	23	26	27	25/27	24/27
27	26	26	29	31	29	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Corcym — Bicarbon™ Family brochure CC-MK-00002 A — <https://www.corcym.com/devices/aortic>
- Own data of the hospital research unit (ICICOR).

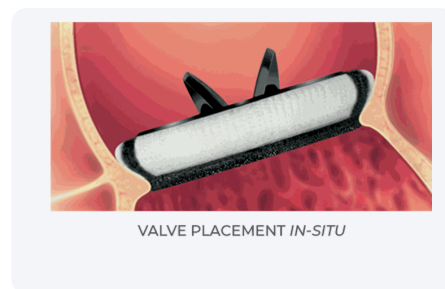
Fitline Aortic



Sorin Bicarbon Fitline aortic — In-situ placement · © Corcym



Sorin Bicarbon Fitline aortic — Photo · © Corcym



Sorin Bicarbon Fitline aortic — In-situ placement · © Corcym



Sorin Bicarbon 1990 — Photo and fluoroscopy

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- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm²)	EOA (cm²)	OH (mm)
19	ICV0917	19.0	15.2	1.76	0.97	6.0
21	ICV0918	21.2	17.2	2.27	1.54	6.4
23	ICV0919	23.4	19.2	2.83	2.07	6.8
25	ICV0920	25.6	21.3	3.45	2.39	7.2
27	ICV0921	27.8	23.3	4.14	3.06	7.6
29	ICV0922	30.0	25.6	5.00	3.45	8.0
31	ICV0923	32.0	25.6	5.00	3.45	8.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20/21.5	20	23	23	23	21
25	21.5	23	26	23	25	24
27	23/24.5	23	26	27	25/27	24/27
29	26	26	29	31	29	27
31	26	26	29	31	29	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Corcym — Bicarbon™ Family brochure CC-MK-00002 A — <https://www.corcym.com/devices/aortic>
- Own data of the hospital research unit (ICICOR).

Fitline Mitral Mitral



Sorin Bicarbon Fitline mitral — In-situ placement · © Corcym



Sorin Bicarbon Fitline mitral — Photo · © Corcym



Sorin Bicarbon Fitline mitral — In-situ placement · © Corcym



Sorin Bicarbon 1990 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- The ring and leaflets are radiopaque.
- Frontal view: its distinctive characteristic is a curved profile (which separates the valve orifice into three equivalent sections).
- Side view: pathognomonic hazy appearance of the curved leaflets.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm²)	OH (mm)
19	ICV0924	19.0	15.2	1.76	6.0
21	ICV0925	21.2	17.2	2.27	6.4
23	ICV0926	23.4	19.2	2.83	6.8
25	ICV0927	25.6	21.3	3.45	7.2
27	ICV0928	27.8	23.3	4.14	7.6
29	ICV0929	30.0	25.6	5.00	8.0
31	ICV0930	32.0	25.6	5.00	8.0
33	ICV0931	34.0	25.6	5.00	8.0

No TAVI sizing table for this model.

SOURCES

- Corcym — Bicarbon™ Family brochure CC-MK-00002 A — <https://www.corcym.com/devices/aortic>

Manufacturer contact: Corcym S.r.l. · communication@corcym.com · Via Giovanni Spadolini 7, Centro Leoni, 20141 Milano, Italy · <https://www.corcym.com/contact-us>

CarboMedics

Corcym (formerly LivaNova / Sorin)

Standard Aortic Aortic



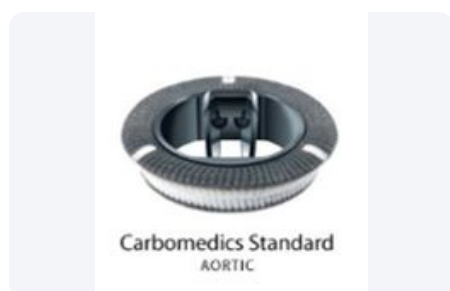
CarboMedics aortic (1996) — Photo · Smithsonian NMAH (CC0)



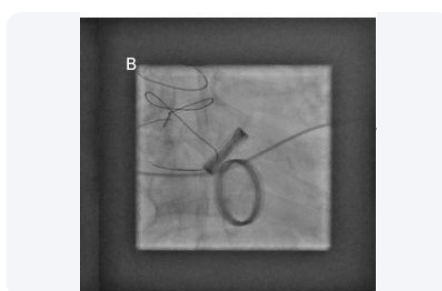
CarboMedics aortic (1996) — Oblique photo · Smithsonian NMAH (CC0)



CarboMedics Standard aortic — Photo · © Corcym



CarboMedics Standard aortic — Catalogue photo · © Corcym



CarboMedics — Fluoroscopy (overview)



CarboMedics 21 mm — Fluoroscopy LAO16 CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavited pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
19	R5-019	18.8	14.7	1.59	1.00	6.2
21	R5-021	20.8	16.7	2.07	1.50	6.6
23	R5-023	22.6	18.5	2.56	1.60	7.3
25	R5-025	25.0	20.5	3.16	2.00	7.7
27	R5-027	27.0	22.5	3.84	2.40	8.4
29	R5-029	29.0	24.2	4.44	2.60	8.7

TAVI SIZE SELECTION

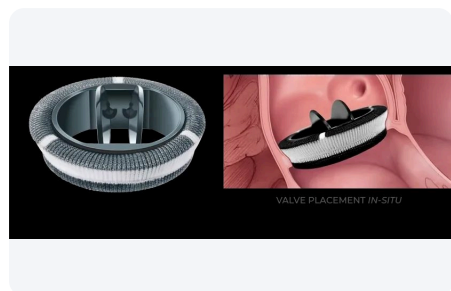
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
21	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
23	▲ 20	▲ 20	23	23	▲ Caution	21
25	20/21.5	20	26	23	23	24
27	23	23	26	27	25	24
29	24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

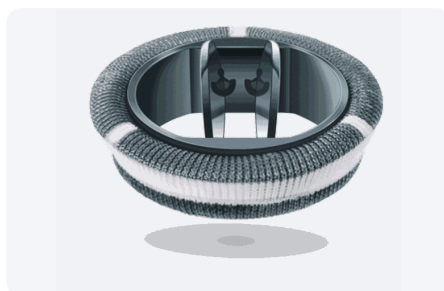
SOURCES

- Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>
- Own data of the hospital research unit (ICICOR).

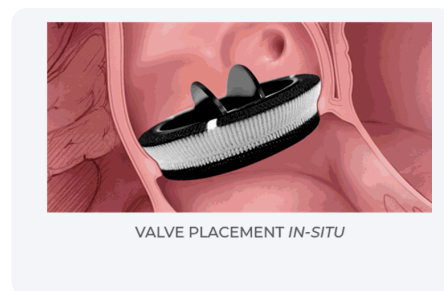
Reduced Aortic Aortic



CarboMedics Reduced aortic — In-situ placement
© Corcym



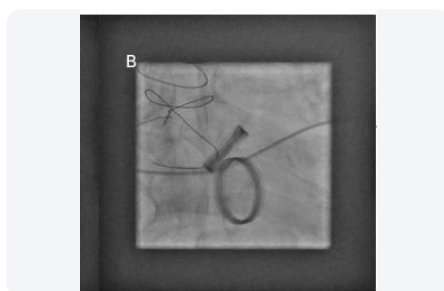
CarboMedics Reduced aortic — Photo · © Corcym



CarboMedics Reduced aortic — In-situ placement
© Corcym



CarboMedics Reduced aortic — Catalogue photo · © Corcym



CarboMedics — Fluoroscopy (overview)



CarboMedics 21 mm — Fluoroscopy LAO16 CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavitated pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
19	A5-019	19.8	14.7	1.59	1.00	6.2
21	A5-021	21.8	16.7	2.07	1.50	6.6
23	A5-023	23.8	18.5	2.56	1.60	7.3
25	A5-025	25.8	20.5	3.16	2.00	7.7
27	A5-027	27.8	22.5	3.84	2.40	8.4
29	A5-029	29.8	24.2	4.44	2.60	8.7
31	A5-031	31.8	24.2	4.44	2.60	8.7

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegro	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
23	⚠ 20	⚠ 20	23	23	⚠ Caution	21
25	20/21.5	20	26	23	23	24
27	23	23	26	27	25	24
29	24.5	26	29	27	27	27
31	24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

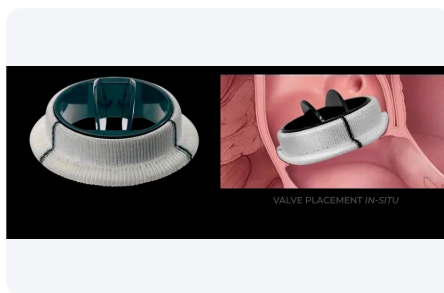
SOURCES

- Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>
- Own data of the hospital research unit (ICICOR).

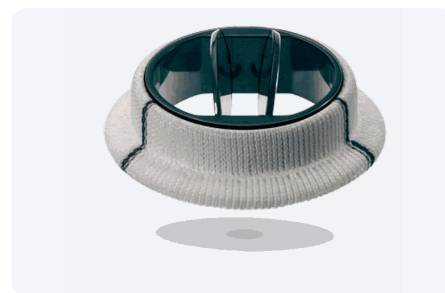
Top Hat Aortic (totally supra-annular) Aortic



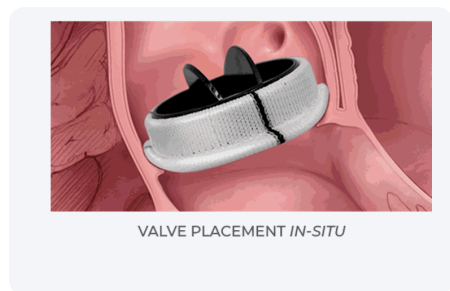
CarboMedics Top Hat aortic — Photo · © Corcym



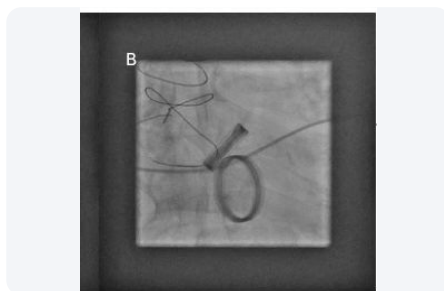
CarboMedics Top Hat aortic — In-situ placement · © Corcym



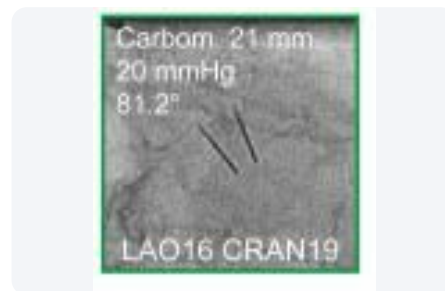
CarboMedics Top Hat aortic — Photo · © Corcym



CarboMedics Top Hat aortic — In-situ placement · © Corcym



CarboMedics — Fluoroscopy (overview)



CarboMedics 21 mm — Fluoroscopy LAO16 CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavitied pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
19	S5-019	14.7	14.7	1.59	1.00	6.2
21	S5-021	16.7	16.7	2.07	1.40	6.6
23	S5-023	18.5	18.5	2.56	1.90	7.3
25	S5-025	20.5	20.5	3.16	2.20	7.7
27	S5-027	22.5	22.5	3.84	2.90	8.4

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
21	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
23	▲ 20	▲ 20	23	23	▲ Caution	21
25	20/21.5	20	26	23	23	24
27	23	23	26	27	25	24

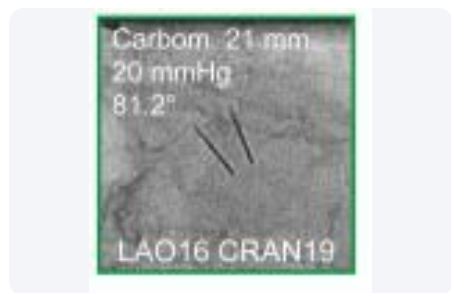
Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>
- Own data of the hospital research unit (ICICOR).

Orbis Aortic Aortic





CarboMedics 21 mm — Fluoroscopy LAO16
CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavitied pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	EOA (cm ²)	OH (mm)
19	A1-019	18.8	14.7	1.59	1.00	6.2
21	A1-021	20.8	16.7	2.07	1.50	6.6
23	A1-023	22.6	18.5	2.56	1.60	7.3
25	A1-025	25.0	20.5	3.16	2.00	7.7
27	A1-027	27.0	22.5	3.84	2.40	8.4
29	A1-029	29.0	24.2	4.44	2.60	8.7
31	A1-031	31.0	24.2	4.44	2.60	8.7

TAVI SIZE SELECTION

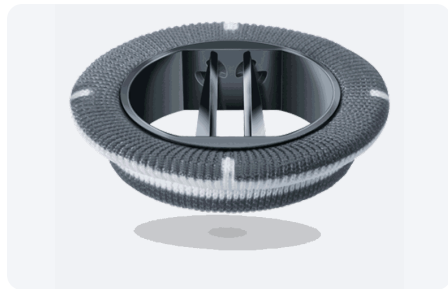
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
21	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
23	▲ 20	▲ 20	23	23	▲ Caution	21
25	20/21.5	20	26	23	23	24
27	23	23	26	27	25	24
29	24.5	26	29	27	27	27
31	24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

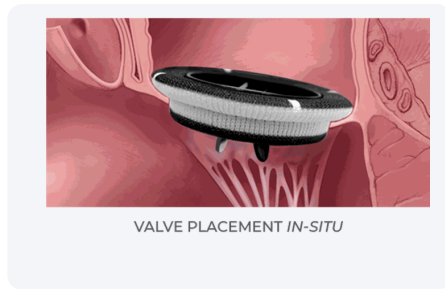
SOURCES

- Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>
- Own data of the hospital research unit (ICICOR).

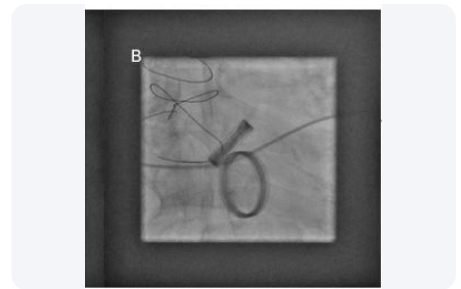
Standard Mitral Mitral



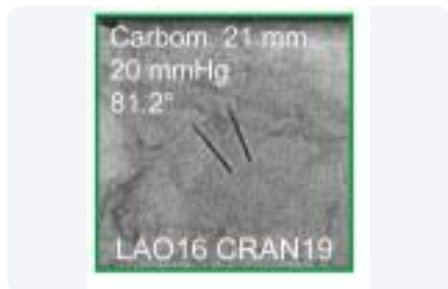
CarboMedics Standard mitral — Photo · © Corcym



CarboMedics Standard mitral — In-situ placement · © Corcym



CarboMedics — Fluoroscopy (overview)



CarboMedics 21 mm — Fluoroscopy LAO16 CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavited pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

TECHNICAL SPECIFICATIONS

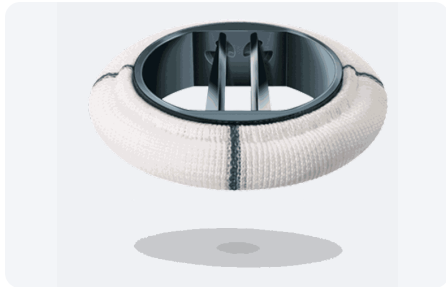
Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	OH (mm)
16	M7-016	16.2	14.7	1.59	6.2
18	M7-018	18.8	14.7	1.59	6.2
21	M7-021	21.8	16.7	2.07	6.6
23	M7-023	23.8	18.5	2.56	7.3
25	M7-025	25.8	20.5	3.16	7.7
27	M7-027	27.8	22.5	3.84	8.4
29	M7-029	29.8	24.2	4.44	8.7
31	M7-031	31.8	24.2	4.44	8.7
33	M7-033	33.8	24.2	4.44	8.7

No TAVI sizing table for this model.

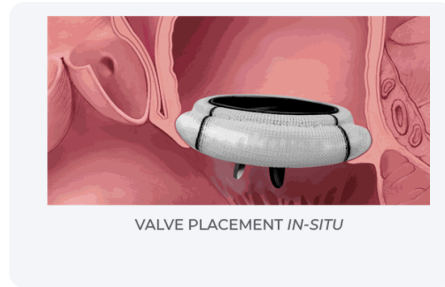
SOURCES

Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>

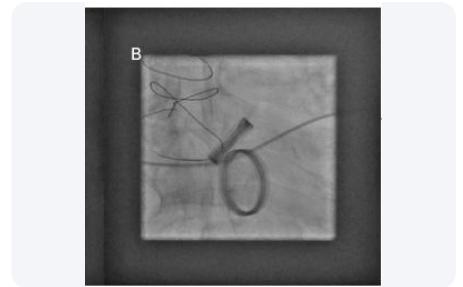
Optiform Mitral Mitral



CarboMedics Optiform mitral — Photo · © Corcym



CarboMedics Optiform mitral — In-situ placement
· © Corcym



CarboMedics — Fluoroscopy (overview)



CarboMedics 21 mm — Fluoroscopy LAO16
CRAN19

FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavited pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

TECHNICAL SPECIFICATIONS

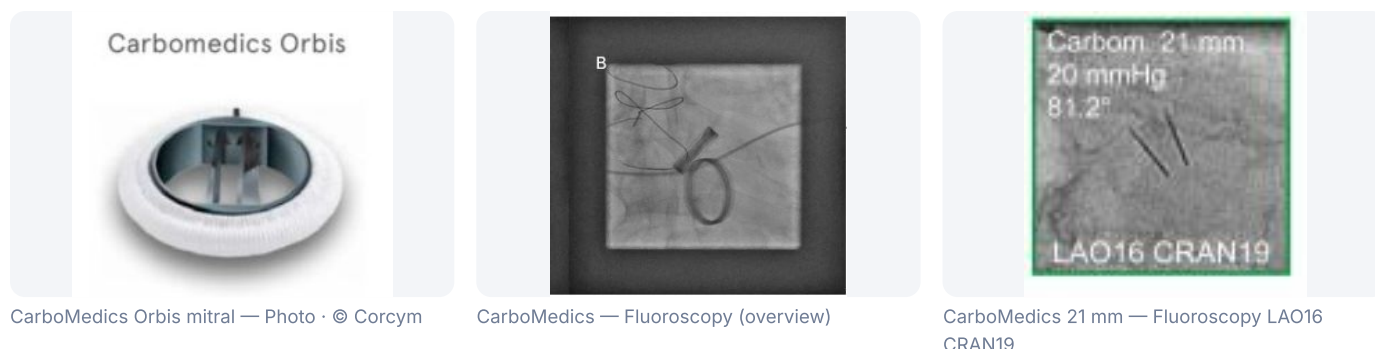
Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	OH (mm)
23	F7-023	22.6	18.5	2.56	7.3
25	F7-025	25.0	20.5	3.16	7.7
27	F7-027	27.0	22.5	3.84	8.4
29	F7-029	29.0	24.2	4.44	8.7
31	F7-031	31.0	24.2	4.44	8.7
33	F7-033	33.0	24.2	4.44	8.7

No TAVI sizing table for this model.

SOURCES

- Corcym — Carbomedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>

Orbis Mitral Mitral



FLUOROSCOPIC IDENTIFICATION

- Has a titanium housing ring reinforced with another titanium ring of greater radiopacity, which allows rotation of the valve for optimal orientation after implantation.
- The cavitied pivot area is embedded inside a pyrolytic carbon housing into which the valve hinges are recessed.
- Edge-to-edge diameter is longer than the hinge-to-hinge diameter because this pivot design alters the circular shape of the valve orifice.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	OH (mm)
21	M2-021	20.8	16.7	2.07	6.6
23	M2-023	22.6	18.5	2.56	7.3
25	M2-025	25.0	20.5	3.16	7.7
27	M2-027	27.0	22.5	3.84	8.4
29	M2-029	29.0	24.2	4.44	8.7
31	M2-031	31.0	24.2	4.44	8.7
33	M2-033	33.0	24.2	4.44	8.7

No TAVI sizing table for this model.

SOURCES

- Corcym — CarboMedics™ Family brochure CC-MK-00003 A (2021) — <https://www.corcym.com/devices/aortic/US/carbo-medics>

Manufacturer contact: Corcym S.r.l. · communication@corcym.com · Via Giovanni Spadolini 7, Centro Leoni, 20141 Milano, Italy · <https://www.corcym.com/contact-us>

Standard Aortic

Aortic



Edwards Duromedics Standard aortic — Photo

TECHNICAL SPECIFICATIONS

Size (mm)	TAD (mm)	ID (mm)	GOA (cm²)	Overall height (mm)
19	19.0	15.0	1.77	7.0
21	21.0	17.0	2.27	7.5
23	23.0	19.0	2.84	8.0
25	25.0	21.0	3.46	8.5
27	27.0	23.0	4.15	9.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20	20	23	23	23	21
25	21.5	23	26	23/27	23/25	24
27	23/24.5	23	26/29	27	25/27	24

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Own data of the hospital research unit (ICICOR).

Standard Mitral

Mitral

TECHNICAL SPECIFICATIONS

Size (mm)	TAD (mm)	ID (mm)	GOA (cm²)	Overall height (mm)
19	19.0	15.0	1.77	7.0
21	21.0	17.0	2.27	7.5
23	23.0	19.0	2.84	8.0
25	25.0	21.0	3.46	8.5
27	27.0	23.0	4.15	9.0

No TAVI sizing table for this model.

SOURCES

■ Own data of the hospital research unit (ICICOR).

Manufacturer contact: Edwards Lifesciences Corp. · +1 949 250 2500 · One Edwards Way, Irvine, CA 92614, USA · <https://www.edwards.com/about-us/global-locations>

Standard Aortic

Aortic



HP-Biplus — Photo and fluoroscopy

TECHNICAL SPECIFICATIONS

Size (mm)	TAD (mm)	ID (mm)	GOA (cm²)	Overall height (mm)
19	19.0	15.2	1.81	6.7
21	21.0	17.2	2.32	7.1
23	23.0	19.2	2.90	7.6
25	25.0	21.2	3.53	8.1
27	27.0	23.2	4.23	8.6

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20	20	23	23	23	21
25	21.5	23	26	27	25	24
27	24.5	23	29	27	27	27

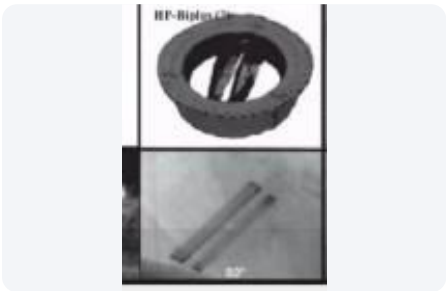
Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Own data of the hospital research unit (ICICOR).

HP-Biplus C

Aortic



HP-Biplus — Photo and fluoroscopy

TECHNICAL SPECIFICATIONS

Size (mm)	TAD (mm)	ID (mm)	GOA (cm²)	Overall height (mm)
19	19.0	15.5	1.89	6.5
21	21.0	17.5	2.41	7.0
23	23.0	19.5	2.99	7.5
25	25.0	21.5	3.63	8.0
27	27.0	23.5	4.34	8.5

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20	20	23	23	23	21
25	21.5/23	23	26	27	25	24
27	23/24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Own data of the hospital research unit (ICICOR).

Standard Aortic

Aortic



Jyros aortic — Photo · Smithsonian NMAH (CC0)



Jyros aortic — Photo, leaflets closed · Smithsonian NMAH (CC0)



Jyros aortic — Oblique photo · Smithsonian NMAH (CC0)



Jyros — Photo and fluoroscopy

TECHNICAL SPECIFICATIONS

Size (mm)	TAD (mm)	ID (mm)	GOA (cm ²)	Overall height (mm)
19	19.0	15.0	1.77	6.5
21	21.0	17.0	2.27	7.0
23	23.0	19.0	2.84	7.5
25	25.0	21.0	3.46	8.0
27	27.0	23.0	4.15	8.5

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20	20	23	23	23	21
25	21.5	23	26	23/27	23/25	24
27	23/24.5	23	26/29	27	25/27	24

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

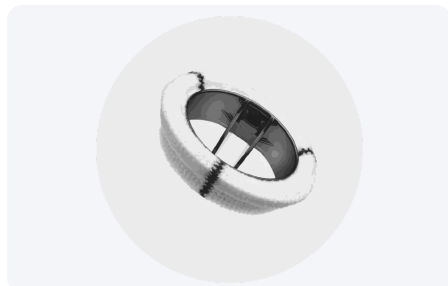
SOURCES

- Smithsonian National Museum of American History — Jyros Bileaflet Heart Valve (object nmah_1726314) — https://www.si.edu/object/jyros-bileaflet-heart-valve:nmah_1726314
- Own data of the hospital research unit (ICICOR).

Medtronic Open Pivot™ (ATS)

Medtronic (formerly ATS Medical)

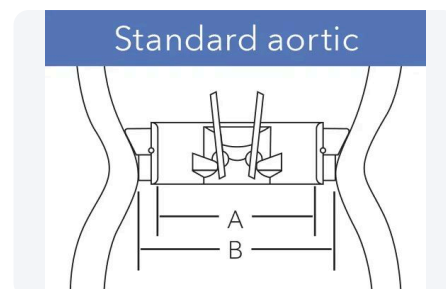
Open Pivot™ Standard Aortic (intra-annular) Aortic



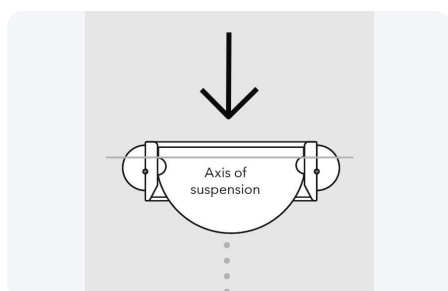
Medtronic Open Pivot Standard aortic — Photo · © Medtronic



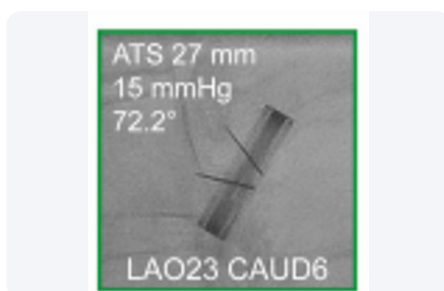
Medtronic Open Pivot — Photo · © Medtronic



Medtronic Open Pivot Standard aortic — Dimensions diagram (A = orifice ID, B = tissue annulus) · © Medtronic



Medtronic Open Pivot — Schematic (pivot axis) · © Medtronic



Medtronic ATS 27 mm — Fluoroscopy LAO23 CAUD6



Medtronic ATS aortic — Fluoroscopy (85° opening)

FLUOROSCOPIC IDENTIFICATION

- Appearance similar to that of the CarboMedics with a few differences.
- Leaflets have increased visibility due to the high tungsten content of the disc substrate.
- Flat titanium ring profile of the ATS valve: similar to some CM models; however, the titanium rotation ring has greater radiopacity.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Overall profile (mm)	Inflow profile (mm)	Outflow profile (mm)
19	500FA19	19.2	14.8	1.55	—	—	—
21	500FA21	21.2	16.8	2.02	10.3	2.9	7.4
23	500FA23	23.2	18.8	2.56	11.3	3.4	7.9
25	500FA25	25.2	20.8	3.17	12.3	3.1	9.2
27	500FA27	27.2	22.8	3.84	13.3	3.4	9.9
29	500FA29	29.2	24.8	4.59	14.3	3.9	10.4
31	500FA31	31.2	26.8	5.35	—	—	—

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution

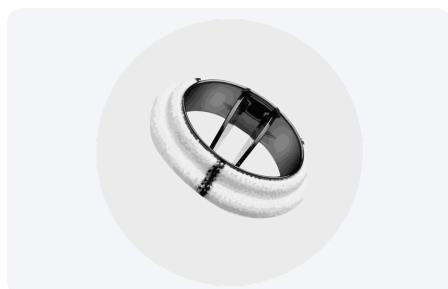
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
21	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
23	20/21.5	20	23	23	23	21
25	20/21.5	20/23	26	23	23	24
27	23	23	26	27	25	24/26
29	24.5/26	26	29	31	27	27
31	27.5	29	34	31	29	30

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

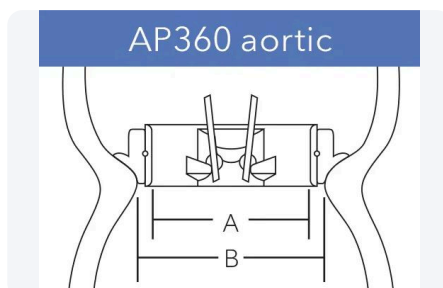
SOURCES

- Medtronic — Open Pivot™ product pages (Standard and AP360™, aortic and mitral): tissue annulus and orifice inner diameter — <https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>
- ATS Medical — Open Pivot® IFU, Table 1: Valve Dimensions (profile heights, inside/outside diameter, orifice area)
- ATS Medical — ATS Open Pivot® Heart Valve Product Guide OPP0012R0; Medtronic Open Pivot™ UC201101694a (2011) — <https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>
- Own data of the hospital research unit (ICICOR).

Open Pivot™ AP / AP360™ Aortic (supra-annular) Aortic



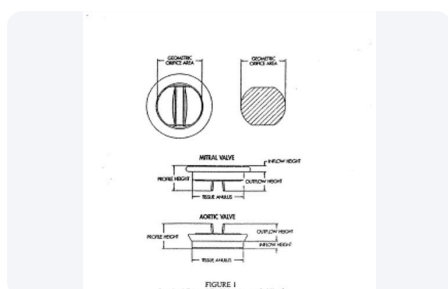
Medtronic Open Pivot AP360 aortic — Photo · © Medtronic



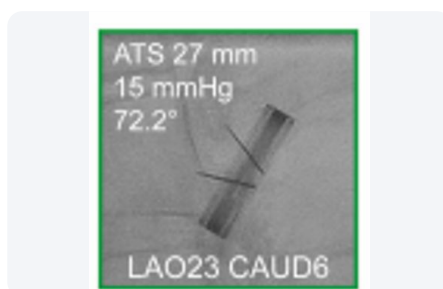
Medtronic Open Pivot AP360 aortic — Dimensions diagram (A = orifice ID, B = tissue annulus) · © Medtronic



Medtronic Open Pivot AP360 — Photo · © Medtronic



Medtronic Open Pivot Standard / AP — Dimensions diagram (orifice area and heights) · © Medtronic



Medtronic ATS 27 mm — Fluoroscopy LAO23 CAUD6



Medtronic ATS aortic — Fluoroscopy (85° opening)

FLUOROSCOPIC IDENTIFICATION

- Appearance similar to that of the CarboMedics with a few differences.
- Leaflets have increased visibility due to the high tungsten content of the disc substrate.
- Flat titanium ring profile of the ATS valve: similar to some CM models; however, the titanium rotation ring has greater radiopacity.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.

- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Overall profile (mm)	Inflow profile (mm)	Outflow profile (mm)
16	505DA16	16.2	14.8	1.55	—	—	—
18	505DA18	18.2	16.8	2.02	10.3	1.5	8.8
20	505DA20	20.2	18.8	2.56	11.3	1.5	9.8
22	505DA22	22.2	20.8	3.17	12.3	1.5	10.8
24	505DA24	24.2	22.8	3.84	13.3	1.5	11.8
26	505DA26	26.2	24.8	4.59	14.3	1.5	12.8
28	505DA28	28.2	26.8	5.35	—	—	—

TAVI SIZE SELECTION

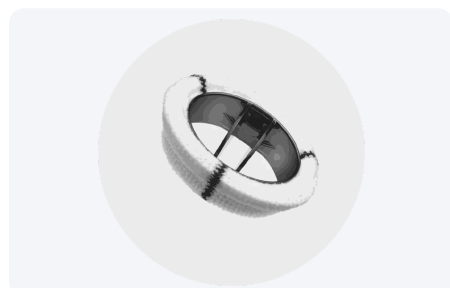
Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
16	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
18	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
20	20/21.5	20	23	23	23	21
22	20/21.5	20/23	26	23	23	24
24	23	23	26	27	25	24/27
26	24.5/26	26	29	31	27	27
28	27.5	29	34	31	29	30

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

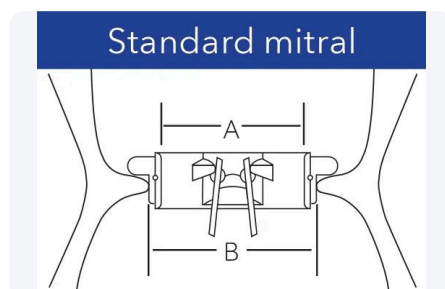
SOURCES

- Medtronic — Open Pivot™ product pages (Standard and AP360™, aortic and mitral): tissue annulus and orifice inner diameter — <https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>
- ATS Medical — Open Pivot® IFU, Table 1: Valve Dimensions (profile heights, inside/outside diameter, orifice area)
- Own data of the hospital research unit (ICICOR).

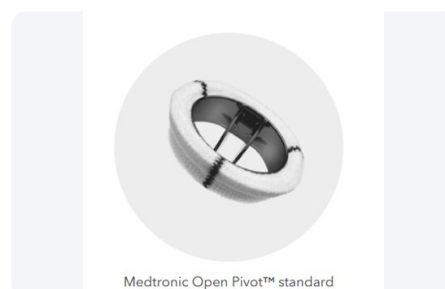
Open Pivot™ Standard Mitral Mitral



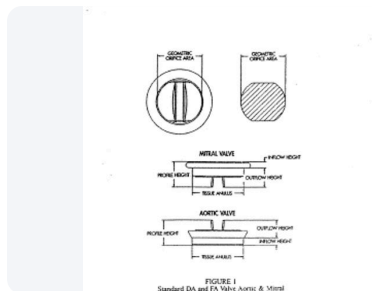
Medtronic Open Pivot Standard aortic — Photo · © Medtronic



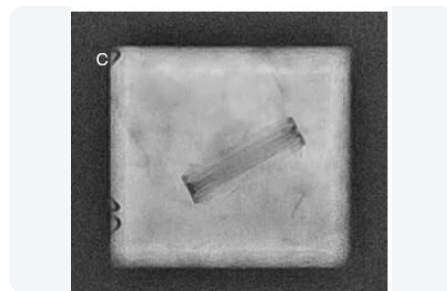
Medtronic Open Pivot Standard mitral — Dimensions diagram (A = orifice ID, B = tissue annulus) · © Medtronic



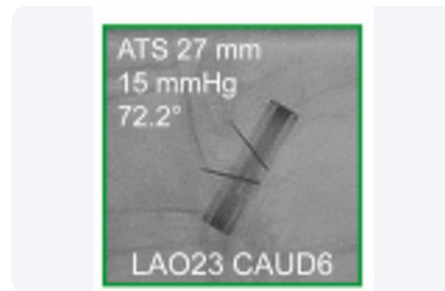
Medtronic Open Pivot Standard — Photo · © Medtronic



Medtronic Open Pivot Standard / AP — Dimensions diagram (orifice area and heights) · © Medtronic



Medtronic ATS — Fluoroscopy (overview)



Medtronic ATS 27 mm — Fluoroscopy LAO23 CAUD6

FLUOROSCOPIC IDENTIFICATION

- Appearance similar to that of the CarboMedics with a few differences.
- Leaflets have increased visibility due to the high tungsten content of the disc substrate.
- Flat titanium ring profile of the ATS valve: similar to some CM models; however, the titanium rotation ring has greater radiopacity.

TECHNICAL SPECIFICATIONS

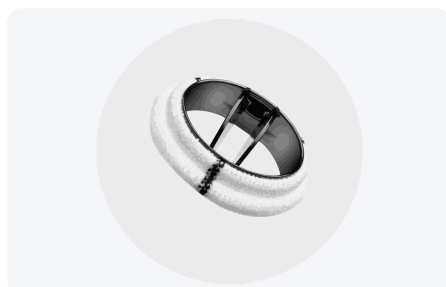
Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Overall profile (mm)	Inflow profile (mm)	Outflow profile (mm)
19	500DM19	19.2	14.8	1.55	—	—	—
21	500DM21	21.2	16.8	2.02	—	—	—
23	500DM23	23.2	18.8	2.56	—	—	—
25	500DM25	25.2	20.8	3.17	—	—	—
27	500DM27	27.2	22.8	3.84	—	—	—
29	500DM29	29.2	24.8	4.59	14.3	3.7	10.6
31	500DM31	31.2	26.8	5.35	15.4	3.7	11.7
33	500DM33	33.2	26.8	5.35	15.4	3.7	11.7

No TAVI sizing table for this model.

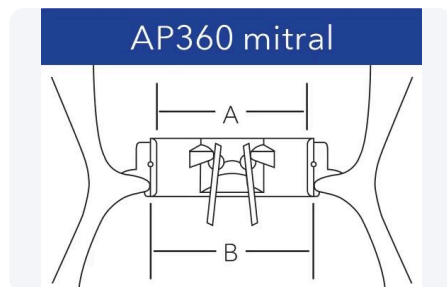
SOURCES

- Medtronic — Open Pivot™ product pages (Standard and AP360™, aortic and mitral): tissue annulus and orifice inner diameter — <https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>
- ATS Medical — Open Pivot® IFU, Table 1: Valve Dimensions (profile heights, inside/outside diameter, orifice area)

Open Pivot™ AP360™ Mitral (supra-annular) Mitral



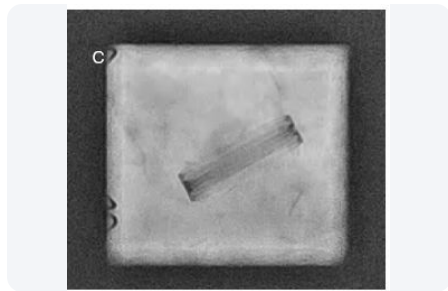
Medtronic Open Pivot AP360 aortic — Photo · © Medtronic



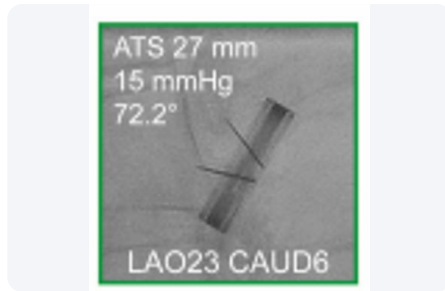
Medtronic Open Pivot AP360 mitral — Dimensions diagram (A = orifice ID, B = tissue annulus) · © Medtronic



Medtronic Open Pivot AP360 — Photo · © Medtronic



Medtronic ATS — Fluoroscopy (overview)



Medtronic ATS 27 mm — Fluoroscopy LAO23 CAUD6

FLUOROSCOPIC IDENTIFICATION

- Appearance similar to that of the CarboMedics with a few differences.
- Leaflets have increased visibility due to the high tungsten content of the disc substrate.
- Flat titanium ring profile of the ATS valve: similar to some CM models; however, the titanium rotation ring has greater radiopacity.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Overall profile (mm)	Inflow profile (mm)	Outflow profile (mm)
16	505DM16	16.2	14.8	1.55	—	—	—
18	505DM18	18.2	16.8	2.02	—	—	—
20	505DM20	20.2	18.8	2.56	—	—	—
22	505DM22	22.2	20.8	3.17	—	—	—
24	505DM24	24.2	22.8	3.84	—	—	—
26	505DM26	26.2	24.8	4.59	14.3	3.7	10.6
28	505DM28	28.2	26.8	5.35	15.4	3.7	11.7

No TAVI sizing table for this model.

SOURCES

- Medtronic — Open Pivot™ product pages (Standard and AP360™, aortic and mitral): tissue annulus and orifice inner diameter — <https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>
- ATS Medical — Open Pivot® IFU, Table 1: Valve Dimensions (profile heights, inside/outside diameter, orifice area)

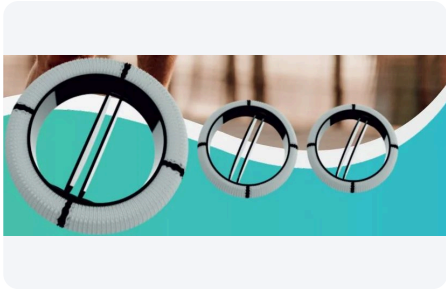
Manufacturer contact: Medtronic plc (ATS Medical) · +1 763 514 4000 · 710 Medtronic Parkway, Minneapolis, MN 55432-5604, USA · <https://www.medtronic.com/en-us/contact-us.html>

Miltonia™ Aortic

Aortic



Meril Miltonia — Photo · © Meril



Meril Miltonia — Several sizes (photo) · © Meril



Meril Miltonia (titanium ring and pivots) — 3D render · © Meril



Meril Miltonia aortic — Several sizes (photo) · © Meril

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm²)	Overall height (mm)	Sewing ring OD (mm)
19	MLT19A	19.0	14.8	1.55	8.7	19.8
21	MLT21A	21.0	16.8	2.02	9.7	21.8
23	MLT23A	23.0	18.8	2.56	10.9	23.8
25	MLT25A	25.0	20.8	3.17	12.1	25.8
27	MLT27A	27.0	22.8	3.84	13.1	27.8

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
23	20	20	23	23	23	21
25	21.5/23	23	26	27	25	24
27	23/24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

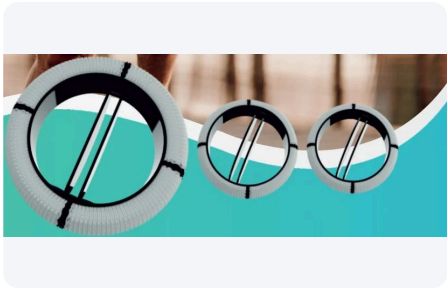
SOURCES

- Meril Life Sciences — MILTONIA™ Bi-Leaflet Mechanical Valve brochure MLT/BROCHURE/003/MLS/20231120/GLOBAL (Ordering information; implant-height table) — https://strapi.merillife.com/uploads/miltonia_1702448206pdf_32c38dec34.pdf
- Own data of the hospital research unit (ICICOR).

Miltonia™ Aortic Advanced Performance (AP) Aortic



Meril Miltonia — Photo · © Meril



Meril Miltonia — Several sizes (photo) · © Meril



Meril Miltonia (titanium ring and pivots) — 3D render · © Meril



Meril Miltonia aortic — Several sizes (photo) · © Meril

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
16	MLT16AP	16.2	14.8	1.55	8.7	8.7	19.8
18	MLT18AP	18.2	16.8	2.02	9.7	9.7	21.8
20	MLT20AP	20.2	18.8	2.56	—	10.9	23.8

No TAVI sizing table for this model.

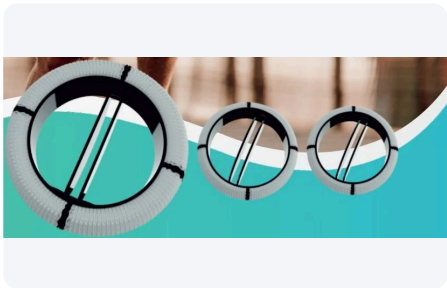
SOURCES

Meril Life Sciences — MILTONIA™ Bi-Leaflet Mechanical Valve brochure MLT/BROCHURE/003/MLS/20231120/GLOBAL (Ordering information; implant-height table) — https://strapi.merillife.com/uploads/miltonia_1702448206pdf_32c38dec34.pdf

Miltonia™ Mitral Mitral



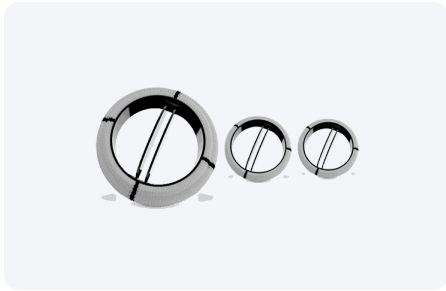
Meril Miltonia — Photo · © Meril



Meril Miltonia — Several sizes (photo) · © Meril



Meril Miltonia (titanium ring and pivots) — 3D render · © Meril



Meril Miltonia aortic — Several sizes (photo) · © Meril

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
23	MLT23M	23.8	18.8	2.56	6.7	11.5	31.0
25	MLT25M	25.8	20.8	3.17	7.1	12.5	33.0
27	MLT27M	27.8	22.8	3.84	7.9	13.5	35.0
29	MLT29M	29.8	24.8	4.59	8.9	14.4	37.0
31	MLT31M	31.8	26.8	5.37	8.9	14.4	39.0
33	MLT33M	33.8	26.8	5.37	9.1	15.4	41.0

No TAVI sizing table for this model.

SOURCES

Meril Life Sciences — MILTONIA™ Bi-Leaflet Mechanical Valve brochure MLT/BROCHURE/003/MLS/20231120/GLOBAL (Ordering information; implant-height table) — https://strapi.merillife.com/uploads/miltonia_1702448206pdf_32c38dec34.pdf

Manufacturer contact: Meril Life Sciences Pvt. Ltd. · +91 260 350 9400, +91 260 240 8000 · askinfo@merillife.com

· Meril Corporate Office, Near Bilakhia Stadium, Muktanand Marg, Chala, Vapi 396 191, Gujarat, India · <https://www.merillife.com/contact-us>

Conform-X® sewing ring Aortic



On-X Conform-X aortic — Photo · © Artivion



On-X aortic — Photo · © Artivion



On-X aortic — Side photo · © Artivion



On-X housing, flared inlet and pivot guards — Detail photo · © Artivion



On-X 1996 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- Orifice: graphite substrate coated with On-X® Carbon (a pure unalloyed form of pyrolytic carbon).
- Leaflets: On-X® Carbon deposited on a graphite substrate impregnated with 10 weight% tungsten to provide radiopacity.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
19	ONXACE-19	19.0	17.4	2.28	27	10.8	13.3
21	ONXACE-21	21.0	19.4	2.84	30	11.9	14.7
23	ONXACE-23	23.0	21.4	3.44	33	13.1	16.1
25	ONXACE-25	25.0	23.4	4.11	34	14.2	17.8
27/29	ONXACE-27/29	27–29	23.4	4.11	36	14.2	17.8

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	23/27	23/25	24
25	23/24.5	23	26/29	27	25/27	24/27

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
27	23/24.5	23	26/29	27	25/27	24/27
29	23/24.5	23	26/29	27	25/27	24/27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 (06/2017) — <https://www.onxlti.com/wp-content/uploads/2019/05/010122103A-On-X-IFU-Booklet-English-Website-Version.pdf>
- Own data of the hospital research unit (ICICOR).

Anatomic sewing ring Aortic



On-X Anatomic aortic — Photo · © Artivion



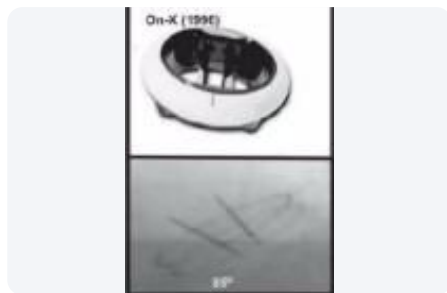
On-X aortic — Photo · © Artivion



On-X aortic — Side photo · © Artivion



On-X housing, flared inlet and pivot guards — Detail photo · © Artivion



On-X 1996 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- Orifice: graphite substrate coated with On-X® Carbon (a pure unalloyed form of pyrolytic carbon).
- Leaflets: On-X® Carbon deposited on a graphite substrate impregnated with 10 weight% tungsten to provide radiopacity.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
19	ONXANE-19	19.0	17.4	2.28	27	10.8	13.3
21	ONXANE-21	21.0	19.4	2.84	30	11.9	14.7
23	ONXANE-23	23.0	21.4	3.44	33	13.1	16.1
25	ONXANE-25	25.0	23.4	4.11	34	14.2	17.8
27/29	ONXANE-27/29	27-29	23.4	4.11	36	14.2	17.8

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	23/27	23/25	24
25	23/24.5	23	26/29	27	25/27	24/27
27	23/24.5	23	26/29	27	25/27	24/27
29	23/24.5	23	26/29	27	25/27	24/27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 (06/2017) — <https://www.onxlti.com/wp-content/uploads/2019/05/010122103A-On-X-IFU-Booklet-English-Website-Version.pdf>
- Own data of the hospital research unit (ICICOR).

Standard sewing ring Aortic



On-X Standard aortic — Photo · © Artivion



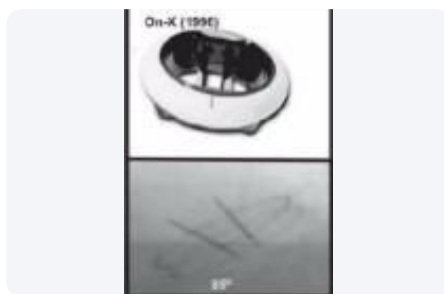
On-X aortic — Photo · © Artivion



On-X aortic — Side photo · © Artivion



On-X housing, flared inlet and pivot guards — Detail photo · © Artivion



On-X 1996 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- Orifice: graphite substrate coated with On-X® Carbon (a pure unalloyed form of pyrolytic carbon).
- Leaflets: On-X® Carbon deposited on a graphite substrate impregnated with 10 weight% tungsten to provide radiopacity.

POSITIONING & DEPLOYMENT

- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
19	ONXAE-19	19.0	17.4	2.28	23	10.8	13.3

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
21	ONXAE-21	21.0	19.4	2.84	26	11.9	14.7
23	ONXAE-23	23.0	21.4	3.44	29	13.1	16.1
25	ONXAE-25	25.0	23.4	4.11	32	14.2	17.8
27/29	ONXAE-27/29	27-29	23.4	4.11	34	14.2	17.8

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	23/27	23/25	24
25	23/24.5	23	26/29	27	25/27	24/27
27	23/24.5	23	26/29	27	25/27	24/27
29	23/24.5	23	26/29	27	25/27	24/27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 (06/2017) — <https://www.onxlti.com/wp-content/uploads/2019/05/010122103A-On-X-IFU-Booklet-English-Website-Version.pdf>
- Own data of the hospital research unit (ICICOR).

On-X® Mitral (standard ring) Mitral



On-X mitral — Photo · © Artivion



On-X mitral — Schematic (design features) · © Artivion



On-X mitral — Photo · © Artivion



On-X 1996 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- Orifice: graphite substrate coated with On-X® Carbon (a pure unalloyed form of pyrolytic carbon).
- Leaflets: On-X® Carbon deposited on a graphite substrate impregnated with 10 weight% tungsten to provide radiopacity.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
23	ONXM-23	23.0	21.4	3.44	31	13.1	16.1
25	ONXM-25	25.0	23.4	4.11	33	14.2	17.8
27/29	ONXM-27/29	27-29	23.4	4.11	34	14.2	17.8
31/33	ONXM-31/33	31-33	23.4	4.11	36	14.2	17.8

No TAVI sizing table for this model.

SOURCES

■ On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 (06/2017) — <https://www.onxlti.com/wp-content/uploads/2019/05/010122103A-On-X-IFU-Booklet-English-Website-Version.pdf>

On-X® Mitral Conform-X® Mitral



On-X mitral — Photo · © Artivion



On-X mitral — Schematic (design features) · © Artivion



On-X mitral — Photo · © Artivion



On-X 1996 — Photo and fluoroscopy

FLUOROSCOPIC IDENTIFICATION

- Orifice: graphite substrate coated with On-X® Carbon (a pure unalloyed form of pyrolytic carbon).
- Leaflets: On-X® Carbon deposited on a graphite substrate impregnated with 10 weight% tungsten to provide radiopacity.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	Orifice area (cm ²)	Sewing ring OD (mm)	Profile h closed (mm)	Profile H open (mm)
25/33	ONXMC-25/33	25-33	23.4	4.11	39	14.2	17.8

No TAVI sizing table for this model.

SOURCES

■ On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 (06/2017) — <https://www.onxlti.com/wp-content/uploads/2019/05/010122103A-On-X-IFU-Booklet-English-Website-Version.pdf>

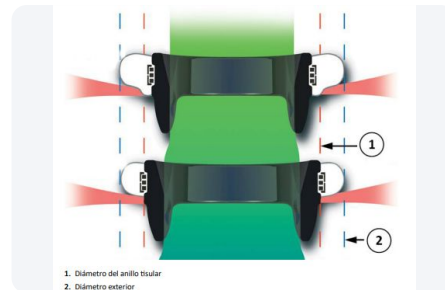
Manufacturer contact: Artivion, Inc. (On-X®) · +1 770 419 3355, +1 800 438 8285 · 1655 Roberts Blvd NW, Kennesaw, GA 30144, USA, On-X Life Technologies: 1300 East Anderson Lane, Bldg B, Austin, TX 78752, USA · <https://artivion.com/contact/>

SJM Regent™ — Standard Cuff **Aortic**

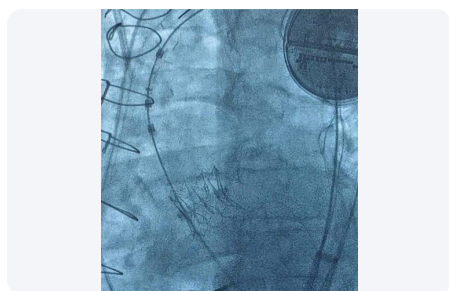
SJM Regent Standard Cuff — Photo · © Abbott



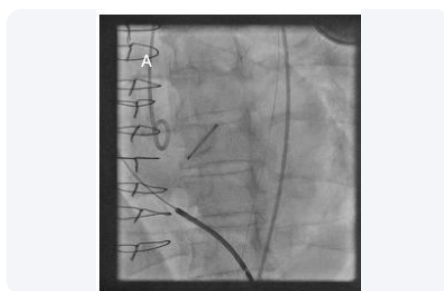
SJM Regent aortic 23 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters — Dimensions diagram (tissue annulus / outer diameter)



SJM Regent aortic — Fluoroscopy



SJM — Fluoroscopy (overview)

FLUOROSCOPIC IDENTIFICATION

- Ring is not radiopaque (most distinguishable feature).
- Only the discs are seen, moving like 'butterfly wings' in the side view.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
17	17AGN-751	17.0	—	1.90	6.1	11.6	24.7
19	19AGN-751	19.0	17.8	2.40	6.1	11.6	24.7
21	21AGN-751	21.0	19.6	2.90	7.3	13.0	26.7
23	23AGN-751	23.0	21.4	3.50	8.0	14.3	28.7
25	25AGN-751	25.0	23.0	4.00	8.2	14.6	30.7
27	27AGN-751	27.0	24.9	4.70	9.1	15.4	32.7

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegro	Navitor	VitaFlow
17	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
19	⚠ Caution	⚠ Caution	⚠ Caution	23	⚠ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	27	25	24

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
25	23	23	26	27	25	24
27	24.5/26	26	29	31	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Abbott — Regent™ Aortic Mechanical Heart Valve — product page, ordering information — <https://www.cardiovascular.abbott/us/en/hcp/products/structural-heart/surgical-valve-solutions/regent-valve.html>
- St. Jude Medical — Regent™ FlexCuff ordering information (2013) — valve orifice inner diameter — https://hcuvicicor.github.io/ViMech_app/assets/SJM_Regent/regent-mechanical-heart-valve-flexcuff.pdf
- Own data of the hospital research unit (ICICOR).

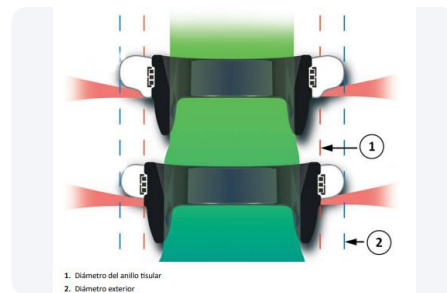
SJM Regent™ — FlexCuff Aortic



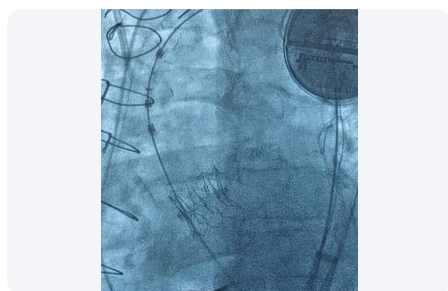
SJM Regent FlexCuff — Photo · © Abbott



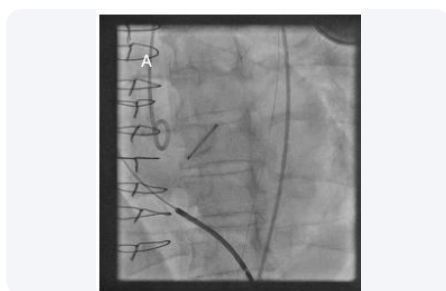
SJM Regent aortic 23 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters — Dimensions diagram (tissue annulus / outer diameter)



SJM Regent aortic — Fluoroscopy



SJM — Fluoroscopy (overview)

FLUOROSCOPIC IDENTIFICATION

- Ring is not radiopaque (most distinguishable feature).
- Only the discs are seen, moving like 'butterfly wings' in the side view.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
17	17AGFN-756	17.0	—	1.90	6.1	11.6	24.7
19	19AGFN-756	19.0	17.8	2.40	6.1	11.6	27.0
21	21AGFN-756	21.0	19.6	2.90	7.3	13.0	29.0

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
23	23AGFN-756	23.0	21.4	3.50	8.0	14.3	31.0
25	25AGFN-756	25.0	23.0	4.00	8.2	14.6	33.0
27	27AGFN-756	27.0	24.9	4.70	9.1	15.4	35.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
17	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
19	▲ Caution	▲ Caution	▲ Caution	23	▲ Caution	21
21	20/21.5	20	23	23	23	21
23	21.5	23	26	27	25	24
25	23	23	26	27	25	24
27	24.5/26	26	29	31	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Abbott — Regent™ Aortic Mechanical Heart Valve — product page, ordering information — <https://www.cardiovascular.abbott/us/en/hcp/products/structural-heart/surgical-valve-solutions/regent-valve.html>
- St. Jude Medical — Regent™ FlexCuff ordering information (2013) — valve orifice inner diameter — https://hcuvicor.github.io/ViMech_app/assets/SJM_Regent/regent-mechanical-heart-valve-flexcuff.pdf
- Own data of the hospital research unit (ICICOR).

SJM Masters™ — HP Series Aortic



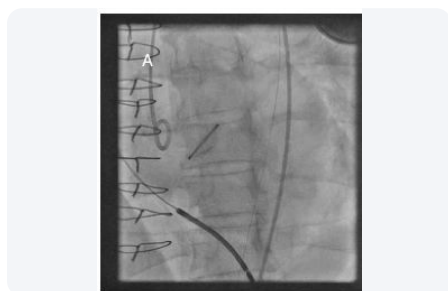
SJM Masters HP aortic 23 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters HP aortic — Photo · © Abbott



SJM Masters HP aortic and mitral — Profile view · © Abbott



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

FLUOROSCOPIC IDENTIFICATION

- Have a radiopaque ring.

- Disks remain suspended further away from the ring's plane because, in contrast to other bileaflet valves, it has a higher pivot point.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
17	17AHPJ-505	17.0	—	1.60	4.6	9.8	22.0/22.5
19	19AHPJ-505	19.0	16.6	2.10	5.2	10.6	24.0/24.0
21	21AHPJ-505	21.0	18.6	2.60	6.0	12.1	26.0/26.5
23	23AHPJ-505	23.0	20.4	3.10	6.2	13.3	28.5/29.0
25	25AHPJ-505	25.0	22.4	3.80	6.7	13.9	30.5/30.5
27	27AHPJ-505	27.0	24.2	4.40	7.5	14.6	32.5/33.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
17	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
19	▲ Caution	▲ Caution	▲ Caution	▲ 23	▲ Caution	▲ Caution
21	▲ 20/21.5	▲ 20	23	23	▲ 23	21
23	20/21.5	20	23/26	23	23	21/24
25	23	23	26	27	25	24
27	24.5	26	29	27	27	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf
- Chambers JB, Oo L, Narracott A, et al. Nominal size in six bileaflet mechanical aortic valves: a comparison of orifice size and biologic equivalence. J Thorac Cardiovasc Surg. 2003;125(6):1388-93 (Table 1, measured inside diameter) — [https://doi.org/10.1016/S0022-5223\(03\)00039-4](https://doi.org/10.1016/S0022-5223(03)00039-4)
- Own data of the hospital research unit (ICICOR).

SJM Masters™ — Polyester Cuff Aortic



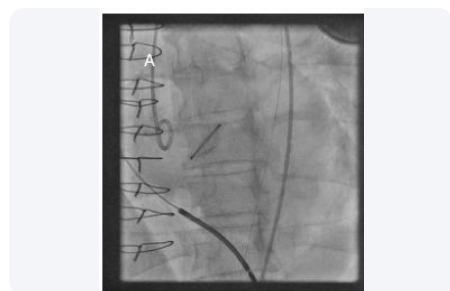
SJM Masters aortic 27 mm (polyester cuff) — Photo · Smithsonian NMAH (CC0)



SJM Masters Standard Cuff — Photo · © Abbott



SJM Masters Expanded Cuff — Photo · © Abbott



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

FLUOROSCOPIC IDENTIFICATION

- Have a radiopaque ring.
- Disks remain suspended further away from the ring's plane because, in contrast to other bileaflet valves, it has a higher pivot point.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
19	19AJ-501	19.0	14.8	1.60	3.5	9.8	22.5/23.0
21	21AJ-501	21.0	16.6	2.10	3.9	10.6	24.5/24.5
23	23AJ-501	23.0	18.6	2.60	5.2	12.1	27.0/28.0
25	25AJ-501	25.0	20.4	3.10	5.5	13.3	29.5/30.0
27	27AJ-501	27.0	22.5	3.80	5.8	13.9	31.0/32.0
29	29AJ-501	29.0	—	4.40	6.5	14.6	33.5/34.0
31	31AJ-501	31.0	—	5.10	7.3	16.0	35.5/37.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegro	Navitor	VitaFlow
19	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution	⚠ Caution
21	⚠ Caution	⚠ Caution	⚠ Caution	⚠ 23	⚠ Caution	⚠ Caution
23	⚠ 20/21.5	⚠ 20	23	23	⚠ 23	21
25	20/21.5	20	23/26	23	23	21
27	23	23	26	27	25	24

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
29	24.5	26	29	27	27	27
31	26/27.5	26	29	31	29	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf
- Chambers JB, Oo L, Narracott A, et al. Nominal size in six bileaflet mechanical aortic valves: a comparison of orifice size and biologic equivalence. J Thorac Cardiovasc Surg. 2003;125(6):1388-93 (Table 1, measured inside diameter) — [https://doi.org/10.1016/S0022-5223\(03\)00039-4](https://doi.org/10.1016/S0022-5223(03)00039-4)
- Own data of the hospital research unit (ICICOR).

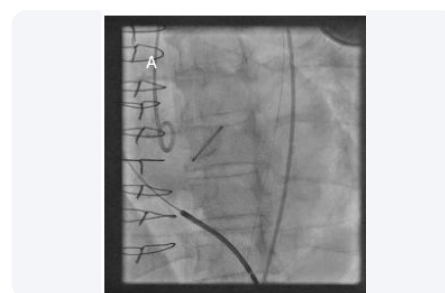
SJM Masters™ — PTFE Cuff Aortic



SJM Masters Standard Cuff — Photo · © Abbott



SJM Masters aortic 27 mm (polyester cuff) — Photo · Smithsonian NMAH (CC0)



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

FLUOROSCOPIC IDENTIFICATION

- Have a radiopaque ring.
- Disks remain suspended further away from the ring's plane because, in contrast to other bileaflet valves, it has a higher pivot point.

POSITIONING & DEPLOYMENT

- Avoid rapid pacing to allow disc to embolise to aortic side.
- With BE valve: target position 80/20 aortic/ventricular.
- With SE valve: target depth up to 3 mm into the ventricle.

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
19	19ATJ-503	19.0	14.8	1.60	3.5	9.8	24.0
21	21ATJ-503	21.0	16.6	2.10	3.9	10.6	26.0
23	23ATJ-503	23.0	18.6	2.60	5.2	12.1	28.5

Size (mm)	Model / code	TAD (mm)	ID (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
25	25ATJ-503	25.0	20.4	3.10	5.5	13.3	30.5
27	27ATJ-503	27.0	22.5	3.80	5.8	13.9	32.5
29	29ATJ-503	29.0	—	4.40	6.5	14.6	35.0
31	31ATJ-503	31.0	—	5.10	7.3	16.0	37.0

TAVI SIZE SELECTION

Size	Myval	SAPIEN 3	Evolut	Allegra	Navitor	VitaFlow
19	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution	▲ Caution
21	▲ Caution	▲ Caution	▲ Caution	▲ 23	▲ Caution	▲ Caution
23	▲ 20/21.5	▲ 20	23	23	▲ 23	21
25	20/21.5	20	23/26	23	23	21
27	23	23	26	27	25	24
29	24.5	26	29	27	27	27
31	26/27.5	26	29	31	29	27

Source: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)

SOURCES

- Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf
- Chambers JB, Oo L, Narracott A, et al. Nominal size in six bileaflet mechanical aortic valves: a comparison of orifice size and biologic equivalence. J Thorac Cardiovasc Surg. 2003;125(6):1388-93 (Table 1, measured inside diameter) — [https://doi.org/10.1016/S0022-5223\(03\)00039-4](https://doi.org/10.1016/S0022-5223(03)00039-4)
- Own data of the hospital research unit (ICICOR).

SJM Masters™ — Polyester Cuff (mitral) Mitral



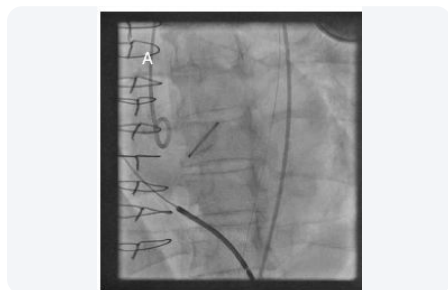
SJM Masters mitral — Photo · Smithsonian NMAH (CC0)



SJM Masters mitral 31 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters Standard Cuff — Photo · © Abbott



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
19	19MJ-501	19.0	1.60	3.8	9.8	23.0/24.0
21	21MJ-501	21.0	2.10	3.7	10.6	25.0/25.5
23	23MJ-501	23.0	2.60	4.6	12.1	28.0/28.5
25	25MJ-501	25.0	3.10	5.2	13.3	30.0/30.5
27	27MJ-501	27.0	3.80	5.8	13.9	32.0/32.0
29	29MJ-501	29.0	4.40	6.7	14.6	34.5/35.0
31	31MJ-501	31.0	5.10	7.1	16.0	36.5/37.0
33	33MJ-501	33.0	5.10	8.3	16.0	38.0/38.5

No TAVI sizing table for this model.

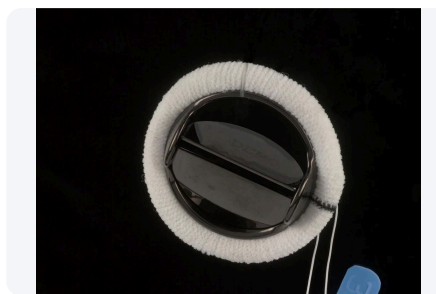
SOURCES

● Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf

SJM Masters™ — HP Series (mitral) Mitral



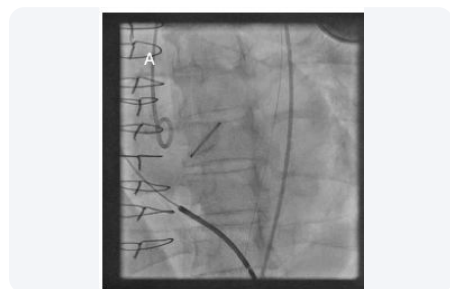
SJM Masters mitral — Photo · Smithsonian NMAH (CC0)



SJM Masters mitral 31 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters HP aortic and mitral — Profile view · © Abbott



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
17	17MHPJ-505	17.0	1.60	2.4	9.8	22.0
19	19MHPJ-505	19.0	2.10	2.7	10.6	24.0
21	21MHPJ-505	21.0	2.60	3.0	12.1	26.0
23	23MHPJ-505	23.0	3.10	3.2	13.3	28.5
25	25MHPJ-505	25.0	3.80	3.8	13.9	30.5
27	27MHPJ-505	27.0	4.40	4.6	14.6	32.0

No TAVI sizing table for this model.

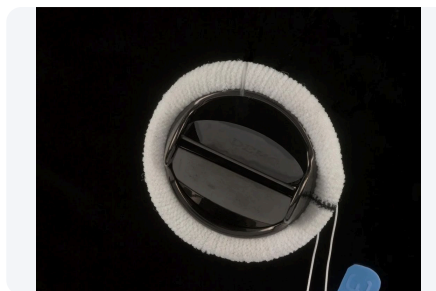
SOURCES

■ Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf

SJM Masters™ — PTFE Cuff (mitral) Mitral



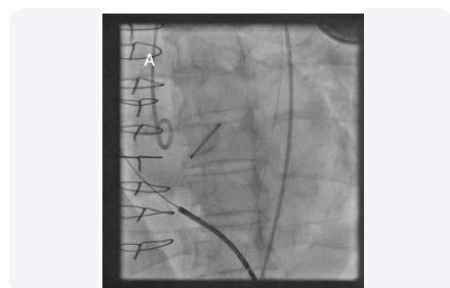
SJM Masters mitral — Photo · Smithsonian NMAH (CC0)



SJM Masters mitral 31 mm — Photo · Smithsonian NMAH (CC0)



SJM Masters Standard Cuff — Photo · © Abbott



SJM — Fluoroscopy (overview)



SJM 23 mm — Fluoroscopy LAO28 CAUD9

TECHNICAL SPECIFICATIONS

Size (mm)	Model / code	TAD (mm)	GOA (cm ²)	Implant height (mm)	Overall height (mm)	Sewing ring OD (mm)
19	19MTJ-503	19.0	1.60	3.8	9.8	24.5
21	21MTJ-503	21.0	2.10	3.7	10.6	26.5
23	23MTJ-503	23.0	2.60	4.6	12.1	29.0
25	25MTJ-503	25.0	3.10	5.2	13.3	31.0
27	27MTJ-503	27.0	3.80	5.8	13.9	33.0
29	29MTJ-503	29.0	4.40	6.7	14.6	35.5
31	31MTJ-503	31.0	5.10	7.1	16.0	37.5
33	33MTJ-503	33.0	5.10	8.3	16.0	39.5

No TAVI sizing table for this model.

SOURCES

■ Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions — https://manuals.eifu.abbott/content/dam/av/manuals-eifu/global/AL/en/ARTMT600261317_C.pdf

Manufacturer contact: Abbott (St. Jude Medical™) · +1 224 667 6100 · 100 Abbott Park Road, Abbott Park, Illinois 60064-6400, USA · <https://www.cardiovascular.abbott/us/en/contact-us.html>

5 TAVI devices in the simulation

Diameter = labelled nominal size. Source of each device height:

Device	Sizes (mm)	Height (mm)	Source
SAPIEN 3	20 · 23 · 26 · 29	15.5 · 18 · 20 · 22.5 height from the manufacturer	Edwards SAPIEN 3 / SAPIEN 3 Ultra IFU (DOC-0160200A), Tables 1a–1b, valve height https://eifu.edwards.com/eifu/5970f1a946e0fb00015e5f4c/DOC-0160200A.pdf
Myval	20 · 21.5 · 23 · 24.5 · 26 · 27.5 · 29 · 30.5 · 32	17 · 17.5 · 18 · 18.5 · 19 · 19.5 · 21 · 21 · 21 approximate height (interpolated from the published range)	Meril Myval brochure: frame height 17–21 mm (range; per-size values interpolated) https://strapi.merillife.com/uploads/myval_1712225907pdf_41222f1e2c.pdf
Evolut	23 · 26 · 29 · 34	— · — · — · — height not verified: schematic drawing, not to scale	Medtronic Evolut: frame height not verified — schematic height https://www.medtronic.com/en-gb/healthcare-professionals/products/cardiovascular/heart-valves/transcatheter-systems/evolut-pro-plus-tavi-system.html
Navitor	23 · 25 · 27 · 29 · 35	47 · 48 · 48 · 48 · 47 height from a secondary source (to be verified)	Abbott Navitor stent height (secondary, unofficial copy of Abbott MAT-2402101) — pending verification https://storage.mtender.gov.md/get/2693b008-8156-4337-ad2f-ce70abb444c1-1732000998730
Allegra	23 · 27 · 31	— · — · — height not verified: schematic drawing, not to scale	Allegra: frame height not verified — schematic height https://eurointervention.pcronline.com/article/the-allegra-transcatheter-heart-valve-european-multicentre-experience-with-a-novel-self-expanding-transcatheter-aortic-valve
VitaFlow	21 · 24 · 27 · 30	50 · 50 · 53 · 53 height from the manufacturer	MicroPort VitaFlow Liberty brochure, valve height H https://microport.com/assets/documents/VitaFlow-Liberty-Brochure.PDF

6 Sources, licence, rights and credits

Data sources

Official source

Documents used for the dimensions in this version of the catalogue (data version 2026-10-06):

- [Abbott — Regent™ Aortic Mechanical Heart Valve — product page, ordering information](#) (SJM)
- [St. Jude Medical — Regent™ FlexCuff ordering information \(2013\) — valve orifice inner diameter](#) (SJM)
- [Abbott — Masters Series™ / Masters HP™ Instructions for Use, Table 1 Reference Dimensions](#) (SJM)
- [Corcym — Carbomedics™ Family brochure CC-MK-00003 A \(2021\)](#) (CarboMedics)
- [Medtronic — Open Pivot™ product pages \(Standard and AP360™, aortic and mitral\): tissue annulus and orifice inner diameter](#) (Medtronic Open Pivot™ (ATS))
- [ATS Medical — Open Pivot® IFU, Table 1: Valve Dimensions \(profile heights, inside/outside diameter, orifice area\)](#) (Medtronic Open Pivot™ (ATS))
- [ATS Medical — ATS Open Pivot® Heart Valve Product Guide OPP0012R0; Medtronic Open Pivot™ UC201101694a \(2011\)](#) (Medtronic Open Pivot™ (ATS))
- [On-X® Prosthetic Heart Valve Instructions for Use, doc. 0459 \(06/2017\)](#) (On-X®)
- [Corcym — Bicarbon™ Family brochure CC-MK-00002 A](#) (Bicarbon™)
- [Meril Life Sciences — MILTONIA™ Bi-Leaflet Mechanical Valve brochure MLT/BROCHURE/003/MLS/20231120/GLOBAL \(Ordering information; implant-height table\)](#) (Meril Miltonia™)

Unofficial data (literature, distributors)

- [Chambers JB, Oo L, Narracott A, et al. Nominal size in six bileaflet mechanical aortic valves: a comparison of orifice size and biologic equivalence. J Thorac Cardiovasc Surg. 2003;125\(6\):1388-93 \(Table 1, measured inside diameter\)](#) (SJM)
- [Smithsonian National Museum of American History — Jyros Bileaflet Heart Valve \(object nmah_1726314\)](#) (Jyros)

Hospital own data

- TAVI size tables: ViMech article, Table 2: THV sizing for the available prosthetic valves (team data, manuscript in preparation)
- Brands without an official source (values not verified): Edwards Duromedics, HP-Biplus, Jyros

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ICICOR — Instituto de Ciencias del Corazón, Hospital Clínico Universitario de Valladolid. ViMech: mechanical valves — fluoroscopic identification and design for valve-in-valve procedures (dimensions and aortic TAVI sizing) [web application]. Data version 2026-10-06. https://hcuvicicor.github.io/ViMech_app/ (accessed 2026-10-08).

Data

Dimensions are technical data taken from the manufacturers' documents listed under *Data sources*; values from literature or distributors are marked as such. TAVI sizing tables are the team's own data.

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Corcym S.r.l. · 35 images

<https://www.corcym.com/devices/aortic>

Corcym product pages and brochures (CarboMedics™, Bicarbon™).

- CarboMedics aortic — Fluoroscopy video 1
- CarboMedics aortic — Fluoroscopy video 2
- CarboMedics aortic — Fluoroscopy video 3
- CarboMedics aortic — TAVI positioning video
- Sorin Bicarbon Fitline aortic — In-situ placement
- Sorin Bicarbon Fitline aortic — Photo
- Sorin Bicarbon Fitline mitral — In-situ placement
- Sorin Bicarbon Fitline mitral — Photo
- Sorin Bicarbon Overline aortic — Photo
- Sorin Bicarbon Slimline aortic — In-situ placement
- Sorin Bicarbon Slimline aortic — Photo
- CarboMedics Optiform mitral — In-situ placement
- CarboMedics Optiform mitral — Photo
- CarboMedics Orbis aortic — Photo
- CarboMedics Orbis mitral — Photo
- CarboMedics Reduced aortic — In-situ placement
- CarboMedics Reduced aortic — Photo
- CarboMedics Standard aortic — Photo
- CarboMedics Standard mitral — In-situ placement
- CarboMedics Standard mitral — Photo
- CarboMedics Top Hat aortic — In-situ placement
- CarboMedics Top Hat aortic — Photo
- CarboMedics Orbis aortic — Catalogue photo
- CarboMedics Reduced aortic — Catalogue photo
- CarboMedics Standard aortic — Catalogue photo
- CarboMedics Standard Pediatric aortic — Catalogue photo
- CarboMedics Top Hat aortic — Catalogue photo
- CarboMedics Reduced aortic — In-situ placement

- CarboMedics Top Hat aortic — Photo
- CarboMedics Top Hat aortic — In-situ placement
- Sorin Bicarbon aortic — Photo
- Sorin Bicarbon family — Photo
- Sorin Bicarbon Fitline aortic — In-situ placement
- Sorin Bicarbon Fitline mitral — In-situ placement
- Sorin Bicarbon Slimline aortic — In-situ placement

Medtronic plc / ATS Medical · 11 images

<https://www.medtronic.com/en-us/healthcare-professionals/products/cardiovascular/heart-valves/mechanical-valves-conduits/medtronic-open-pivot-mechanical-heart-valve.html>

Medtronic product pages and ATS Medical / Open Pivot™ documentation.

- Medtronic Open Pivot AP360 — Photo
- Medtronic Open Pivot Standard — Photo
- Medtronic Open Pivot Standard / AP — Dimensions diagram (orifice area and heights)
- Medtronic Open Pivot AP360 aortic — Photo
- Medtronic Open Pivot — Schematic (pivot axis)
- Medtronic Open Pivot — Photo
- Medtronic Open Pivot Standard aortic — Photo
- Medtronic Open Pivot AP360 aortic — Dimensions diagram (A = orifice ID, B = tissue annulus)
- Medtronic Open Pivot AP360 mitral — Dimensions diagram (A = orifice ID, B = tissue annulus)
- Medtronic Open Pivot Standard aortic — Dimensions diagram (A = orifice ID, B = tissue annulus)
- Medtronic Open Pivot Standard mitral — Dimensions diagram (A = orifice ID, B = tissue annulus)

Artivion, Inc. (On-X®) · 12 images

<https://www.onxlti.com>

Artivion / On-X Life Technologies product pages and documentation.

- On-X aortic sewing ring — Photo
- On-X aortic — Photo
- On-X mitral — Photo
- On-X Anatomic aortic — Photo
- On-X Standard aortic — Several sizes (photo)
- On-X Conform-X aortic — Photo
- On-X Standard aortic — Photo
- On-X aortic — Photo
- On-X aortic — Side photo
- On-X housing, flared inlet and pivot guards — Detail photo
- On-X mitral — Photo
- On-X mitral — Schematic (design features)

Meril Life Sciences Pvt. Ltd. · 4 images

https://strapi.merillife.com/uploads/miltonia_1702448206pdf_32c38dec34.pdf

MILTONIA™ brochure MLT/BROCHURE/003/MLS/20231120/GLOBAL.

- Meril Miltonia aortic — Several sizes (photo)
- Meril Miltonia — Photo
- Meril Miltonia — Several sizes (photo)
- Meril Miltonia (titanium ring and pivots) — 3D render

Abbott (St. Jude Medical™) · 10 images

<https://www.cardiovascular.abbott/us/en/hcp/products/structural-heart/surgical-valve-solutions/regent-valve.html>

Abbott / St. Jude Medical product pages and ordering information.

- SJM aortic — Balloon-expandable valve positioning video

- SJM aortic — Self-expanding valve positioning video 1
- SJM aortic — Self-expanding valve positioning video 2
- SJM aortic — Myval implantation video
- SJM Masters Standard Cuff — Photo
- SJM Masters Expanded Cuff — Photo
- SJM Masters HP aortic — Photo
- SJM Masters HP aortic and mitral — Profile view
- SJM Regent FlexCuff — Photo
- SJM Regent Standard Cuff — Photo

Smithsonian National Museum of American History · 13 images · Public-domain dedication (CC0 1.0)

<https://www.si.edu/openaccess>

Open Access collection; object identifier shown for each image.

- SJM Regent aortic 23 mm — Photo — [NMAH-RWS2014-03743](#)
- SJM Masters HP aortic 23 mm — Photo — [NMAH-RWS2014-03745](#)
- SJM Masters aortic 27 mm (polyester cuff) — Photo — [NMAH-RWS2014-03742](#)
- SJM Masters mitral — Photo — [NMAH-RWS2014-03744](#)
- SJM Masters mitral 31 mm — Photo — [NMAH-RWS2014-03741](#)
- CarboMedics aortic (1996) — Photo — [NMAH-AHB2015q060856-000001](#)
- CarboMedics aortic (1996) — Oblique photo — [NMAH-AHB2015q060858-000001](#)
- Jyros aortic — Photo — [NMAH-AHB2015q060912-000001](#)
- Jyros aortic — Photo, leaflets closed — [NMAH-AHB2015q060911-000001](#)
- Jyros aortic — Oblique photo — [NMAH-AHB2015q060909-000001](#)
- Medtronic ATS Open Pivot (1992) — Photo — [NMAH-RWS2016-09489](#)
- On-X early model (Bokros) — Photo — [NMAH-AHB2015q060852-000001](#)
- On-X early model (Bokros) — Oblique photo — [NMAH-AHB2015q060849-000001](#)

Other figures

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About ViMech

ViMech is a quick-reference guide to the fluoroscopic identification and design of mechanical heart valves, intended as a reference for valve-in-mechanical (TAVI in mechanical prosthesis) procedures.

Each model shows the document its dimensions come from. Models marked *unverified* use the hospital's own data because no official source has been located yet.

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